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Board Gender Diversity and Sustainability Reporting Quality:

A Content Analysis of Swedish Large-Cap Industrial Firms, 2016–2023

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ABSTRACT

The aim of this thesis is to explore the correlation between board gender diversity (BGD) and sustainability reporting quality (SRQ) of 15 large-cap industrial companies in Sweden within the timeframe from 2016 to 2023. Although board gender diversity is often presented as one of the key drivers of better corporate sustainability outcomes, empirical evidence suggests that this is not necessarily true and much of the research to-date is based on generic aggregated ESG scores, which may not capture real differences in sustainability reporting quality. This paper tackles that issue by analyzing 120 firm-year observations with a hand-coded measure of SRQ that includes six dimensions: framework adoption, target specificity, disclosure scope, materiality assessment, third-party assurance and adverse-impact disclosure.

The descriptive analysis revealed the SRQ has significantly increased over the time period, more than tripling from a mean of 5.27 in 2016 to 17.54 in 2023, and the BGD has remained relatively stable at 27.1% female board representation in 2016 and 27.9% in 2023. The effect of BGD on SRQ is statistically significant in the absence of year fixed effects, but not when year fixed effects are added, suggesting that the association is driven more by a time effect than by a firm-specific effect, which is most likely a tightening of the sustainability disclosure regulation (the NFRD and CSRD) in the EU. No single SRQ dimension is strongly and independently linked to BGD.

Legitimacy theory, institutional theory and critical mass theory are used as a theoretical framework to interpret these findings. The interpretation that best fits the data is given by legitimacy theory and institutional theory, with the latter emphasizing that the study's specification of a linear model may not be appropriate for the study's findings of a non-linear threshold effect because of the narrow range of BGD in this sample. The thesis makes a methodological contribution by highlighting the fact that the BGD-SRQ association, when estimated without taking into account the shared time trends, can overestimate the effect of board gender diversity on the materiality of the reported sustainability information; it makes a practical one by warning against using board gender diversity as an adequate proxy for the actual progress in the field of sustainability reporting and in the UN Sustainable Development Goal 5. The study's limitations are that it has a small sample size, high correlation among governance measures and is observational and non-causal in design.

Keywords: Board Gender Diversity; Sustainability Reporting Quality; Content Analysis; Legitimacy theory; Institutional theory; Critical mass theory; Corporate governance

SUMMARY

In addition to financial reporting, companies today are now asked to release reports detailing their contribution to the environment and society. Many investors, regulators and researchers believe that a company that has more women on its board makes better sustainability reports, which is one of the reasons the Fung Initiative for Gender in Leadership has been established. The suggestion is frequently advanced as a means to achieve more gender-balanced boards which will bring real change to corporate transparency and accountability.

The thesis directly tested this idea by analyzing 15 large Swedish industrial firms (household names in engineering and manufacturing) and their sustainability reporting in each of the seven years between 2016 and 2023, a total of 120 company-years. This study did not rely on a single overall “sustainability score” like many previous studies, but instead read each year's company-specific sustainability report and scored it on six specific criteria: whether the report was based on an internationally recognized sustainability standard, whether the company made clear, measurable targets, how many different topics on sustainability were covered, whether it explained which issues were most important to the company and not others, whether the report was audited by an external auditor, and importantly, whether the company was honest about its failures and negative impacts, rather than just its successes.

The study also noted the percentage of female representation on each company's board each year to determine if this figure changed alongside reporting quality.

Two clear trends were identified. It is worth noting that the quality of sustainability reporting had increased significantly over the eight years - over three times on the study's scoring scale. Second, there was virtually no change at all in the percentage of women on these boards over the same period of time, the figure increasing by less than one point. An initial correlation between board diversity and reporting quality was found when the two were tested statistically, but this connection was eliminated after the analysis controlled for the fact that the vast majority of companies have enhanced their reporting performance around the same time. The timing is very much in line with the two areas of EU regulation that impose higher standards on sustainability disclosure, effective in the same years. That is, the improvement in reporting appears to have been influenced by new regulations and laws, and not by the share of women on a board.

Another trend to note is that businesses became a lot more adept in articulating what they are doing well and what they are talking about, but remain significantly underdeveloped at the level of acknowledging issues, falling short, or even delivering substandard results. The implication is that there has been no equivalent progress in transparency over the areas in need of improvement.

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Key Terms and Abbreviations

- BGD – Board Gender Diversity: proportion of female directors.
- SRQ – Sustainability Reporting Quality: structured measure of reporting depth and rigor.
- GRI – Global Reporting Initiative: widely used sustainability reporting framework.
- TCFD – Task Force on Climate-related Financial Disclosures: climate risk reporting guidelines.
- CSRD – Corporate Sustainability Reporting Directive: EU regulation expanding disclosure requirements.
- SBTi – Science Based Targets initiative: climate target validation framework.
- NFRD – Non-Financial Reporting Directive: EU directive mandating sustainability disclosures.
- ÅRL – Årsredovisningslagen: Swedish Annual Accounts Act governing non-financial reporting.
- Blau Index – Statistical measure of diversity across categorical variables.
- ROA – Return on Assets: profitability indicator used as control variable.

Chapter 1: Introduction

1.1 Background

In the last decade, questions about the role of corporate boards in influencing sustainability practices have gained prominence in both academic research and policy debate. As regulatory requirements for sustainability disclosure have become stricter across the European Union, most recently in accordance with the Corporate Sustainability Reporting Directive (CSRD), which extends mandatory reporting obligations from 2025, governance structures that produce credible, substantive, and high-quality sustainability reports have come under increasing scrutiny from researchers, policymakers, and institutional investors (Christensen, Hail, & Leuz, 2021; Pantazi, 2024). Among the governance characteristics that have received particular attention in this context, board gender diversity (BGD), typically operationalized as the proportion of female directors on the board, has been the subject of an expanding empirical literature and significant legislative intervention, including the EU Women on Boards Directive adopted in 2022 (Schoonjans, 2024; Bota-Avram, 2025).

The theoretical background to expect the effect of BGD on sustainability reporting quality (SRQ) draws on three complementary mechanisms. Legitimacy theory suggests that gender-diverse boards may be more attentive to the legitimacy signals their organizations communicate, including those conveyed through sustainability reports (as discussed in Bota-Avram, 2025). Institutional theory extends this by explaining why firms operating under shared regulatory and normative pressures tend to converge in their disclosure practices over time, independent of individual board composition (DiMaggio & Powell, 1983). Critical mass theory adds a further consideration: the influence of a minority group such as female directors on board outcomes may only become detectable once representation crosses a critical threshold, rather than increasing smoothly with each additional appointment (Kanter, 1977).

Although theoretically coherent, these arguments are, however, not supported by empirical evidence linking BGD and sustainability outcomes. Several meta-analyses and systematic reviews acknowledge that positive associations are observable in the aggregate but are heavily conditioned by firm type, institutional context, board governance quality, and the specific outcome measure applied (Byron & Post, 2016; Velte, 2019; Bota-Avram, 2025). Research that uses more rigorous methods to address endogeneity, such as instrumental variable approaches or quasi-experimental designs, generally finds weaker or more conditional effects

than the standard ordinary least squares estimation (Sila, Gonzalez, & Hagendorff, 2016). A particular source of inconsistency in this literature is the tendency to use third-party ESG ratings as a proxy of sustainability reporting quality or performance. The pairwise correlations between six major ESG rating agencies range from 0.38 to 0.71, while for credit ratings, it is 0.99 (Berg, Koelbel & Rigobon, 2022). This indicates that significant methodological differences fundamentally constrain the comparability of studies using different rating sources.

This study addresses these limitations by taking Sweden as a context with high baseline BGD and strong sustainability reporting norms, and by measuring SRQ directly through systematic content analysis of primary annual and sustainability reports instead of using third-party ESG scores. We examine the 15 Swedish large-cap industrial firms listed on the OMX Stockholm Large Cap Index for the period 2016–2023, which includes the implementation of the EU Non-Financial Reporting Directive (NFRD) in Sweden and the planning period for CSRD adoption. This design allows for a longitudinal assessment of whether BGD is associated with improvements in SRQ as regulatory disclosure expectations increase over time.

1.2 Problem Statement and Research Gap

This study is inspired and centered around three particular tensions in the existing literature. First, the majority of the previous research that connect BGD to sustainability outcomes use either aggregate ESG scores or individual performance metrics as the dependent variable. ESG scores differ quite significantly among various providers, reflecting differences in scope, measurement methodology and weighting rather than a stable underlying construct, as Berg et al. (2022) demonstrate. Similarly, Chatterji et al. (2016) indicate low inter-rater agreement between social ratings by leading US providers, which implies that ESG scores are an unreliable indicator for sustainability reporting quality in specific cases. In a study of German companies required to report under the EU CSR Directive, Gerwanski, Kordsachia, and Velte (2019) show that content analysis-based measures of sustainability reporting quality identify governance associations that are not visible when ESG proxies are used. The result directly contributes to this research's methodology.

Second, while a number of studies find positive correlations between BGD and particular sustainability disclosure outcomes, such as voluntary climate disclosure (Ben-Amar, Chang, & McIlkenny, 2017), the quality of greenhouse gas disclosure (Liao, Luo, & Tang, 2015), and carbon disclosure (Endo, 2025), comparatively few concentrates on the substantive quality and depth of sustainability reports as their primary outcome. The many large-sample studies use

disclosure indices based on GRI compliance or ESG sub-scores that capture reporting breadth rather than quality. Bota-Avram (2025) identifies this as a key gap in her recent integrative review of the women-on-boards and sustainability reporting literature, calling for studies that examine what firms actually communicate in their sustainability reports rather than whether they produce reports conforming to standard checklists.

Third, the Swedish large-cap industrial context is underrepresented in the empirical literature despite being theoretically important. According to Glass, Cook, and Ingersoll (2016), BGD effects on sustainability outcomes might be less pronounced in high-governance, high-diversity situations where reporting is already a condition that is institutionally anticipated for major Swedish listed companies. This raises a unique and practically important question: Do other institutional measures weaken the marginal contribution of board composition, or does BGD offer governance value in a setting where baseline sustainability disclosure is already strong? This particular study views this as an open empirical topic and consistently takes a critical and impartial position toward the anticipated direction of findings.

1.3 Research Aim and Questions

The purpose of the study is to investigate the association between board gender diversity and the level of quality and depth of sustainability reporting of the big-cap industrial companies in Sweden in the time frame 2016–2023, through systematic content analysis of the annual and sustainability reports. The study does not assume a positive relationship but is empirically open to the research question. Three research questions guide the study:

RQ1. Is board gender diversity positively associated with the quality and depth of sustainability reporting among Swedish large-cap industrial firms?

RQ2. Which specific dimensions of sustainability reporting quality, including target specificity, use of recognized frameworks, third-party verification, and breadth of environmental and social disclosure, vary systematically with board gender composition?

RQ3. How does the relationship between board gender diversity and sustainability reporting quality evolve over the 2016–2023 period, and what does this suggest about the influence of changing regulatory expectations?

1.4 Significance of the Study

This study makes several contributions to the academic literature and to policy debates about board governance and sustainability reporting. Theoretically, it integrates legitimacy theory, institutional theory, and critical mass theory into a multi-mechanism framework that is applied to a specific and well-defined empirical context rather than a broad cross-country comparison. In doing so, it responds to calls in the literature for more contextually grounded and methodologically precise research on the governance-sustainability relationship (Bota-Avram, 2025; Velte, 2019; Rao & Tilt, 2016).

Methodologically, the use of systematic content analysis of primary documents rather than third-party ESG scores addresses a recognised limitation that has been shown to affect the comparability and reliability of findings in this literature (Berg et al., 2022; Chatterji et al., 2016; Gerwanski et al., 2019). The content analysis coding scheme developed for this study is grounded in GRI Standards, TCFD recommendations, and Science-Based Target commitments, providing a transparent and reproducible quality assessment.

From a policy perspective, findings from this study are relevant to ongoing debates about whether board diversity mandates, such as those introduced by the EU Women on Boards Directive, are associated with governance improvements in the domain of sustainability disclosure. If BGD is found to be positively associated with SRQ, this provides targeted empirical support for such policies in the industrial sector specifically. If findings are null or conditional, this is equally informative, suggesting that diversity requirements must be embedded within broader governance reform strategies to produce measurable improvements in reporting quality (Schoonjans, 2024).

1.5 Scope, Delimitations, and Structure

The scope of this study is restricted to the large-cap industrial companies that were listed on the OMX Stockholm Large Cap Index during 2016 – 2023. By focusing on the industrial sector, the study narrows its sample to firms that face the highest direct pressure to report on sustainability among the largest emitters and are the most closely scrutinized by regulators and civil society.

The large-cap focus means that there are regular annual and sustainability reports available to the public throughout the observation period. The 2016-2023 period covers the regulatory environment after the Paris Agreement as well as the preparatory phase for implementation of

the CSRD in Sweden (2017) and the expected regulatory differences in the longitudinal dimension of the analysis. The study is based on documents that were all made publicly available, a limitation because of the fact that it is not possible to observe what is happening inside the firms' governance processes, but also a strength because it is what the firms actually say to external actors that is examined. A critical neutral approach is used throughout. A positive BGD-SRQ association is not the expected or desired finding, but also a null association can provide a theoretically informative finding in the context examined, as pointed out by Sila et al. (2016) and Glass et al. (2016). The thesis is organized as follows: Chapter 2 presents a review of the literature and formulation of the theoretical framework; Chapter 3 details the methodology used to conduct the research; Chapter 4 presents empirical findings; Chapter 5 discusses findings within the theoretical framework and literature; and Chapter 6 provides a summary of the contributions, limitations, and future directions.

Chapter 2: Literature Review

This chapter reviews the literature on the primary research question for the study: Is there a relationship between gender diversity on the board and the quality of sustainability reporting and, if so, what is that relationship? It is a thematic review that begins by examining the nature, goals and constraints of sustainability reporting as an accountability practice; it then considers how the quality of corporate governance shapes the quality of sustainability reporting; it looks at theoretical frameworks that link BGD to governance outcomes; it reviews the empirical evidence relevant to BGD and sustainability outcomes, focusing on both positive and negative results; and it concludes by considering the specific gaps this study addresses in the Swedish institutional context. A critical and balanced approach has been used throughout; the intention is not to establish a case for a preconceived conclusion but to consider carefully what the available evidence does and does not indicate, and where there is an unexplainable lack of clarity.

2.1 Sustainability Reporting: Practices, Quality, and Measurement

According to Christensen et al. (2021) and Pantazi (2024), sustainability reporting has evolved from a voluntary practice used by a small number of significant firms in the early 2000s to an increasingly required institutional obligation across major economies. The primary frameworks used by businesses to organize and disseminate their environmental, social, and governance standards include the Global Reporting Initiative (GRI) Standards, the Task Force on Climate-related Financial Disclosures (TCFD), and, more recently, the European Sustainability Reporting Standards (ESRS) under CSRD. The most consistently significant predictors of reporting adoption and breadth include business size, industry involvement, financial success, and stakeholder pressure, according to Hahn and Kühnen's (2013) systematic assessment of 178 papers on SR determinants. More recent systematic reviews confirm the persistence of these determinants while also identifying the quality of board governance as an increasingly recognized factor (Velte, 2019; Gerwanski et al., 2019).

There is a critical distinction that appears in this literature but is not consistently maintained: the quality of sustainability disclosure is different from its quantity. The existence of standalone sustainability reports, the number of GRI indicators disclosed, or the time frame of the report are all observable and comparable. But they are not adequate measures of content standards. Dhanani and Connolly (2014), Gerwanski et al. (2019), and Traxler, Greiling and Hebesberger

(2020) all identify the specificity and time-boundness of the targets set as defining quality, the scope of issues addressed, the inclusion of adverse impacts in reporting as part of its quality, the presence of third-party assurance, and the depth of engagement with stakeholders in materiality assessments as components of quality. In a study on the use of non-governmental organizations' (NGOs) GRI standards, Traxler et al. (2020) show that materiality analysis can be superficial and non-systematic, even among those NGOs formally using comprehensive reporting frameworks, thereby revealing that the use of GRI standards does not guarantee reporting quality.

Third-party ESG ratings have been the dominant method for measuring SRQ in empirical research and are used to combine data from across the environmental, social and governance spectrum into a composite score. However, the pairwise correlation of the different ESG ratings from the six leading agencies is between 0.38 and 0.71, compared with 0.99 for the credit ratings (Berg, Koelbel & Rigobon 2022). They believe the difference is largely due to measurement and scope differences (56% and 38%, respectively), and weighting (6%). The result is important because it raises questions for research on governance determinants of sustainability: research based on different ESG rating sources can give different results or even contradictory results, not because there is no governance relationship, but because the measure of the effect reflects different constructs. This concern is corroborated by Chatterji et al. (2016), who say there is not a high level of consensus on an evaluation of the social dimensions from six major US providers. In the current study, the use of direct content analysis of primary documents is motivated by the findings of these previous studies, which indicate that SRQ measures based on content analysis can identify governance associations that are not captured by ESG proxies as suggested by Gerwanski et al. (2019).

Evidence is provided by Eccles, Ioannou and Serafeim (2014) that the companies that voluntarily transitioned to sustainability policies over the 1990s not only outperformed matched companies in terms of financial metrics but also on non-financial metrics as well, suggesting a real, and not merely symbolic, role for SR. But Christensen et al. (2021) in their extensive review of the mandatory CSR and sustainability reporting literature find that the overall findings regarding the impact of mandatory disclosure requirements on the sustainability of a company are ambiguous and dependent on the nature of the disclosure requirement and the specific disclosure context. Under certain conditions, in theory, companies may have an incentive to produce only positive environmental information and withhold negative data, a practice known as greenwashing, even in the presence of mandatory disclosure, if the external assurance does not include all negative data, as Lyon and Maxwell (2011)

develop. These tensions between ‘sustainability communication for the real’ and ‘sustainability communication for the show’ are directly relevant to the content analysis approach adopted in this study.

2.2 Corporate Governance and Sustainability Reporting Quality

As the highest level of corporate governance, boards of directors are ultimately in charge of the quality and reliability of public disclosures, including sustainability reports (Ben-Amar et al., 2017; Velte, 2019). Board size, independence, expertise, committee structures, and gender composition have all been studied in empirical studies on board attributes and SR quality. A combination of governance mechanisms, such as board gender diversity, sustainability committees, sustainability-linked executive compensation, and external assurance, significantly predicts higher mandatory SR quality as determined by content analysis, according to Gerwanski et al.'s (2019) analysis of 220 German companies subject to mandatory sustainability reporting. The complementarity of these mechanisms implies that BGD is one of several relevant governance factors and that its effect may depend on whether supporting structures are also in place.

Ben-Amar et al. (2017) find, in a cross-country panel study, that female board representation is a significant positive predictor of corporate engagement with voluntary climate disclosure through the Carbon Disclosure Project, with the effect robust to controls for firm size, financial performance, and industry. Post, Rahman, and Rubow (2015) find positive associations between female board representation combined with sustainability committee presence and environmental governance quality in a US sample. Endo (2025) extends this evidence base to Japan, finding in a panel of 2,169 firm-year observations from 2017 to 2023 that female outside directors significantly enhance carbon disclosure quality, with the effect moderated by board expertise diversity, supporting the resource dependence interpretation that the sustainability-relevant expertise brought by diverse directors is a key conduit for the BGD effect.

However, critical counterpoints must be acknowledged. Walls, Berrone, and Phan (2012) find in a study of US industrial and non-industrial firms that board characteristics explain limited variation in environmental performance and governance outcomes once industry and firm-level controls are included. Sila, Gonzalez, and Hagendorff (2016) use instrumental variable approaches to address endogeneity in a large UK panel and find no significant evidence that BGD influences firm performance across a range of governance outcomes, arguing that

positive associations observed in other studies likely reflect selection rather than causal governance effects. Rao and Tilt (2016), in a study of Australian listed firms, find that sustainability committee presence is a stronger predictor of SR quality than general board diversity, suggesting that targeted governance structures may matter more than overall board composition for disclosure outcomes.

2.3 Theoretical Frameworks

2.3.1 Legitimacy Theory

Legitimacy theory argues that organisations engage in disclosure practices partly to manage their legitimacy to signal to external stakeholders that their activities align with prevailing social and institutional norms (as reviewed in Bota-Avram, 2025; Lyon & Maxwell, 2011). Gender-diverse boards may enhance the legitimacy-signalling function of sustainability reports by providing visible evidence of inclusive governance that is valued by investors, regulators, and civil society, who increasingly attend to both board composition and disclosure quality. Endo (2025) notes that female non-executive directors contribute to sustainability disclosure primarily through monitoring and legitimacy channels rather than direct strategic control, a finding that aligns with this perspective.

Legitimacy theory also generates a more critical reading of the BGD-SRQ relationship. If sustainability reporting primarily serves legitimacy management purposes rather than genuine accountability, then gender-diverse boards may produce reports that appear more comprehensive without underlying improvements in governance or sustainability practices. Lyon and Maxwell (2011) demonstrate theoretically that firms with incentives to project environmental responsibility may engage in selective, impression-managing disclosure that is difficult to distinguish from genuine reporting quality using surface-level indicators. The possibility that observed BGD-SRQ associations reflect legitimacy management rather than substantive governance improvement is a serious alternative interpretation that the study acknowledges as a boundary condition of its findings.

2.3.2 Institutional Theory and Isomorphism

Institutional theory extends legitimacy-based reasoning by specifying the mechanisms through which organizations operating in a shared environment come to resemble one another over time, a process DiMaggio and Powell (1983) term institutional isomorphism. Three isomorphic pressures are identified: coercive isomorphism, arising from formal regulatory or legal

requirements; mimetic isomorphism, arising from firms imitating peers perceived as successful or legitimate under conditions of uncertainty; and normative isomorphism, arising from shared professional norms and standards, such as those promulgated by accounting and sustainability-reporting bodies.

Applied to sustainability reporting, institutional theory predicts that firms operating within the same regulatory jurisdiction and reporting period will tend to converge in their disclosure practices as formal requirements and professional norms tighten, largely independent of firm-specific characteristics such as board composition. This is a more mechanistic account than legitimacy theory alone provides: where legitimacy theory explains why an individual firm is motivated to appear legitimate, institutional theory explains why many firms subject to the same coercive pressure, such as the EU Non-Financial Reporting Directive (NFRD) or the Corporate Sustainability Reporting Directive (CSRD) would be expected to improve their disclosure practices in parallel and at similar points in time, rather than at a pace determined by internal governance change. Institutional theory therefore provides a directly testable complement to legitimacy theory in longitudinal panel settings, where a shared time trend across otherwise heterogeneous firms is itself suggestive evidence of coercive isomorphism.

2.3.3 Critical Mass Theory

Critical mass theory, originating in Kanter's (1977) work on organizational tokenism, proposes that the behavioral and strategic influence of a minority group within a decision-making body is not a linear function of its numerical representation. Where women constitute only a small minority of a board (so-called “token” representation), their influence on board dynamics and outcomes is theorized to be limited, as individual members are more likely to conform to majority norms or be perceived through gendered stereotypes rather than contribute distinct perspectives. Kanter's framework and subsequent extensions of it suggest that a meaningful shift in board dynamics and outcomes requires representation to cross a critical threshold commonly proposed in the governance literature to lie in the region of three members or approximately 30 percent of the board, beyond which minority members are more able to form coalitions, voice distinct viewpoints, and influence collective decision-making, including on non-financial and sustainability matters.

Critical mass theory carries a specific empirical implication that is distinct from the other perspectives discussed in this chapter: it predicts a non-linear, threshold-based relationship between board gender diversity and organizational outcomes, rather than the smooth, linear

relationship assumed by standard regression specifications. Under this account, a board with two of ten female directors (20 percent) and a board with three of ten female directors (30 percent) may differ substantially in their capacity to influence disclosure outcomes, even though the numerical difference in percentage-point representation is modest. This has direct implications for empirical studies, such as the present one, in which sample board gender diversity clusters within a relatively narrow range: if the sample's variation in BGD sits close to, or straddles, the theorized critical mass threshold without extending clearly above or below it, a linear model may be poorly suited to detecting whatever relationship critical mass theory predicts, independent of whether such a relationship exists in the underlying population of firms.

2.3.4 Summary of the Theoretical Framework

This thesis therefore draws on three complementary theoretical perspectives: legitimacy theory and institutional theory, which emphasise the firm's, and firms' collective, response to external social and regulatory pressure; and critical mass theory, which contributes a distinct methodological caution, namely that the relationship between board gender diversity and organisational outcomes may be non-linear and threshold-dependent rather than smoothly continuous. Together, these perspectives provide a framework capable of interpreting both the presence and the absence of a detectable BGD–SRQ association in the empirical analysis that follows.

2.4 Empirical Evidence on BGD and Sustainability Outcomes

2.4.1 Supportive Evidence

There are significant numbers of empirical studies that find positive links between having at least one woman on the board and sustainability outcomes, but such studies differ in the outcome measures, measurement methods and contexts. In a cross-country study, Ben-Amar et al. (2017) discover that boards with increased proportions of female directors are significantly more prone to make voluntary CDP climate disclosures, after adjusting for firm size, financial performance and industry. However, Liao, Luo and Tang (2015) find positive correlations between gender diversity and quality of voluntary disclosure of GHGs in FTSE 350 companies and Lu and Herremans (2019) report similar findings for carbon disclosure quality in a North American sample of firms. The latest direct evidence on disclosure quality comes from Endo (2025), who in his study of 2,169 Japanese firm-years reveals that female outside directors

have a significant positive impact on carbon disclosure, particularly when there is a high level of expertise diversity within the board.

In a broader perspective, Gerwanski et al. (2019) conclude that gender diversity at the supervisory board level, alongside the presence of sustainability committees, sustainability-linked remuneration and external assurance, are significant predictors of quality of mandatory sustainability reporting as measured by content analysis in the sample of German companies. Menicucci and Paolucci (2024) document that a higher share of women on the board is linked to better disclosure quality, measured in GRI, of the Italian listed firms. A meta-analytic review of board characteristics and sustainability reporting by Yadav and Prashar (2022) shows that gender diversity has significant positive mean effects on the quality of reporting on sustainability with high variability. In an international study of non-financial companies, Chang & Wu (2024) report a positive link between the presence of female directors and CSR practices, with the correlation being stronger in countries with stronger institutions. Collectively, these studies provide cross-national support for a positive BGD-SRQ association while also highlighting its context-dependence.

Finally, in relation to this study's context, the evidence suggests that supporting governance structures act as moderators of the BGD–SRQ relationship. The results of Gerwanski et al. (2019) also suggest that the positive impact of BGD on SRQ is stronger when BGD is coupled with other governance mechanisms, suggesting that BGD is not a substitute for other governance structures. In their sample of U.S. companies, Post et al. (2015) also conclude that female representation on the board and the presence of a sustainability committee lead to better environmental governance results than either of the two measures.

2.4.2 Mixed, Null, and Critical Evidence

In addition to the literature that supports this, there is a considerable amount of research that poses a number of important qualifications that cannot be ignored. In a large UK panel, Sila, Gonzalez and Hagendorff (2016) employ instrumental variables methods to overcome the endogeneity problem and conclude that there is no significant evidence of a link between BGD and firm performance or firm governance after accounting for the selection of female directors into firms with already high governance standards. They argue that much of the positive correlation in previous research could be the result of reverse causality: firms that already have high levels of board diversity may already be well governed, rather than diversity itself causing better governance. Therefore, the same causal direction as in the empirical relationship between

sustainability commitment and female director appointment is a relevant endogeneity concern in the Swedish context, as institutional investor stewardship and nomination committee processes could have a positive impact on the former.

A meta-analysis of 87 studies on BGD and corporate social performance by Byron and Post (2016) shows a small but statistically significant mean effect (r = 0.15) but also that there is significant heterogeneity in the results that is only partially explained by the moderating variables. The results of Glass et al. (2016) are directly applicable to the context of large listed Swedish industrial firms where sustainability reporting is both legally required and normatively expected, as the researchers conclude that the BGD effects on the adoption of sustainability elements were lower in these firms. Chang and Wu (2024) show that BGD effects on CSR are weaker in countries with already strong gender equality norms, which may apply to Sweden. Bota-Avram (2025) notes that the literature has not reached consensus on whether BGD influences sustainability outcomes through substantive governance improvement or primarily through legitimacy signalling, and that distinguishing these mechanisms is a key unresolved question.

Behlau et al. (2024), in their systematic review of board diversity measurement, raise a methodological concern relevant to interpreting the existing literature: studies use heterogeneous operationalizations of BGD ranging from the proportion of female directors to binary indicators, Blau indices, and critical mass thresholds, and these choices affect findings in ways that are rarely acknowledged. The choice of how to measure BGD interacts with the choice of how to measure sustainability outcomes, creating substantial scope for publication bias and specification searching in this literature. The present study adopts a transparent and consistent operationalization of BGD as the proportion of female elected directors on the board in each firm-year, as disclosed in corporate governance statements.

2.5 The Swedish Institutional Context

Sweden's institutional environment is relevant to interpreting the expected relationship between BGD and SRQ in this study's sample. On gender equality, Sweden ranks consistently among the highest-performing countries in the EU, and large Swedish listed companies have achieved relatively high levels of female board representation through market and institutional pressure, principally through the Swedish Code of Corporate Governance's recommendation for gender balance without mandatory quotas (Schoonjans, 2024). This distinguishes Sweden from quota-adopting countries such as Norway and France, where quasi-experimental research designs have been used to estimate the causal effects of mandated BGD increases (Schoonjans, 2024;

Barroso et al., 2024). The absence of a Swedish quota means that the present study cannot exploit such a design, but it also means that BGD variation across Swedish firms reflects genuine governance choices rather than regulatory compliance thresholds.

On sustainability reporting, the mandatory non-financial reporting regime established by the EU NFRD and transposed into Swedish law through the Årsredovisningslagen (ÅRL) from 2017 requires large listed companies to disclose information on environmental, social, and employee matters, human rights, and anti-corruption (Pantazi, 2024). However, the adoption of CSRD from 2025, which introduces a double materiality approach, links the disclosure to ESRS, and introduces the requirement for assurance, increases the requirements regarding disclosure for large-cap Swedish companies. A logical way to consider whether BGD is linked to better disclosures is to think about how firms with a more governance-oriented board, such as a more governance-oriented board that has stronger BGD, might adapt more proactively to growing reporting obligations (Gerwanski et al., 2019).

However, the potential ceiling effect must be admitted straight out of the gate. The overall level of sustainability governance in Sweden is high, and large, listed industrial companies could be generating reasonably good sustainability reports regardless of board composition because of compliance with regulations, institutional investor demands and the reputation effects. If the SRQ variation is restricted by this ceiling then a significant BGD association is only statistically possible if an actual governance association exists. In high governance institutional settings, both Glass et al. (2016) and Chang and Wu (2024) report similar ceiling effect problems. This is a recognized boundary condition which is discussed in the methodology and discussion chapter.

2.6 Sustainability Reporting in the Industrial Sector

The research on sustainability reporting has special features with regard to industrial companies. Industrial companies are among the biggest greenhouse gas emitters, and most easily identified targets of regulatory and civil society demands for environmental governance, which makes sustainability reporting more demanding and credible (Eccles et al., 2014; Toppinen & Korhonen-Kurki, 2013). Meanwhile, the complexity of industrial operations, long value-chain and the challenges of measuring and revealing scope 3 emissions pose real challenges to boards in terms of the quality of disclosure (Toppinen & Korhonen-Kurki, 2013).

Several patterns have been documented in research on SR practices in industrial firms (McAllister et al., 1992; Moffat, 1995; Stippler and Gayle, 1996; and Schilling and Dasgupta, 1999). As with the overall determinants of SR literature, larger industrial companies are more likely to implement the GRI standards, require external assurance and to publish quantitative environmental objectives (Hahn & Kühnen, 2013). Content analysis (Endo, 2025) reveals that a larger number of large industrial companies in Sweden have adopted Science Based Targets (SBTs) setting emission reduction targets aligned with a 1.5°C pathway. Despite the fact that a large majority of companies publish full sustainability reports, however, research shows that materiality assessments are often superficial, scope 3 disclosures are incomplete and that the targets are not time bound, but rather aspirational and not verifiable (Traxler et al., 2020; Toppinen & Korhonen-Kurki, 2013). These patterns describe the dimensions for variation of SRQ in the current study's sample.

2.7 Synthesis and Research Gap

The literature reviewed in this chapter supports the existence of theoretical mechanisms through which BGD may be associated with higher sustainability reporting quality, operating primarily through legitimacy management, institutional convergence, and critical-mass dynamics, while also establishing that the empirical evidence is heterogeneous, context-sensitive, and methodologically fragmented. Supportive findings from Ben-Amar et al. (2017), Gerwanski et al. (2019), Liao et al. (2015), Endo (2025), and others suggest that female board representation is associated with higher-quality sustainability disclosures in a range of contexts. Critical and null findings from Sila et al. (2016), Glass et al. (2016), and Walls et al. (2012) establish that this relationship is not universal and is likely conditioned by endogeneity, institutional context, and baseline governance quality. The meta-analytic evidence from Byron and Post (2016) and Velte (2019) confirms that positive effects are detectable in aggregate but substantially moderated and that heterogeneity is the norm rather than the exception.

Three specific gaps define the contribution of the present study. First, most studies in this literature use ESG scores or narrow performance metrics rather than direct content-analysis-based measures of SRQ, a limitation demonstrated empirically by Gerwanski et al. (2019) and validated methodologically by Berg et al. (2022) and Chatterji et al. (2016). The present study addresses this by applying a systematic content analysis scheme to primary documents from 15 Swedish large-cap industrial firms. Second, the Swedish industrial sector is

underrepresented in the empirical literature despite the theoretical salience of its high-governance, high-diversity institutional context. Third, the 2016–2023 longitudinal window captures the full implementation period of the EU NFRD in Sweden and the preparatory phase for CSRD, providing temporal variation in regulatory expectations that has not previously been examined in this context. By addressing these gaps, the present study aims to contribute evidence that is methodologically transparent, contextually grounded, and informative regardless of whether a positive or null relationship between BGD and SRQ is found.

Chapter 3: Methodology

The methodology used in the study is described in this chapter. It starts with an introduction to the philosophical underpinnings of the study and how that guided the research approach, followed by the research design, sample selection and data collection methods. It provides an explanation of the content analysis coding scheme that was applied for the evaluation of the quality of sustainability reporting (SRQ), defines the process for preparing and validating the resulting dataset and outlines the ways in which the coded data was analyzed, before moving on to discuss the issues of trustworthiness, limitations and ethical concerns. The methodological decisions are always explained in light of the research questions and methodological literature at the time.

3.1 Research Philosophy and Approach

This study adopts a mixed-methods research design that combines a qualitative, interpretive stage of document coding with a quantitative, positivist-oriented stage of panel regression analysis. At the coding stage, the study draws on an interpretivist logic: as discussed by Pretorius (2024) and Lim (2024), interpretivism treats meaning as context-dependent and prioritizes close reading and interpretation over the direct application of predetermined statistical categories. This logic is appropriate for coding the qualitative content of sustainability reports, where judgement is required to assess what a firm actually communicates the language, commitments and governance structures it provides to external parties and how these communicative contents vary between firms and over time. These questions require close, interpretive engagement with documentary evidence rather than the application of predetermined statistical models to numerical proxies alone (Lim, 2024).

This interpretive coding logic is particularly well suited to studies of corporate disclosure because, as Dhanani and Connolly (2014) argue, the content and framing of sustainability reports is shaped by organizational values, stakeholder relationships, and strategic positioning features that are qualitatively distinct across organizations and cannot be adequately captured by composite ESG scores or binary disclosure indicators. Once this qualitative coding produces a structured, numerical SRQ score for each firm-year, however, the study shifts to a quantitative, positivist logic to analyze the resulting panel dataset systematically, using descriptive statistics and fixed-effects regression. This two-stage design qualitative interpretation followed by quantitative analysis follows the methodological approach used by

Gerwanski, Kordsachia, and Velte (2019) in their study of German mandatory sustainability reporting and by Manetti and Toccafondi (2014) in their analysis of GRI-compliant reports, both of which combine structured content coding with statistical analysis of the resulting scores.

3.2 Research Design

The study employs a longitudinal, multi-firm document analysis design. Documents are used as primary empirical materials for this study, such as annual reports, sustainability reports, and corporate governance statements. As Lim (2024) notes, documents are valuable sources in content analysis research because they are produced for purposes beyond the research context and therefore reflect genuine organizational communication rather than responses constructed for a researcher. The annual and sustainability reports are the most obvious source of primary data for a study focused on what firms disclose because they are publicly available, and they are the main way that firms communicate their sustainability governance, commitments and performance.

The longitudinal dimension of the design covering eight years from 2016 to 2023 serves two purposes. First, it allows changes in each firm's sustainability reporting practices over time to be traced, providing a within-firm analytical dimension that supplements the cross-firm comparisons. Second, the observation window spans the implementation of the EU Non-Financial Reporting Directive (NFRD) in Sweden in 2017 and the preparatory phase for the Corporate Sustainability Reporting Directive (CSRD), providing natural regulatory variation that may be reflected in reporting quality trajectories across firms (Christensen, Hail, & Leuz, 2021; Pantazi, 2024). This longitudinal dimension aligns with calls in the sustainability reporting literature for research that goes beyond single-year cross-sections to examine how reporting practices evolve in response to institutional pressures (Gerwanski et al., 2019; Traxler, Greiling, & Hebesberger, 2020).

3.3 Sample Selection

The sample size is comprised of 15 large cap industrial stocks from the OMX Stockholm Large Cap Index in the country of Sweden. The selection was made intentionally, that is, the cases selected have characteristics that are relevant to the research questions, rather than statistical representativeness, which is usually obtained by random sampling. Purposive sampling is widely used in document-based research because the aim is to generate analytically rich

evidence from cases with substantive relevance to the phenomenon under investigation (Lim, 2024; Ahmad & Wilkins, 2025).

Three criteria were used. First, companies had to be traded on the OMX Stockholm Large Cap Index for most of the period (2016–2023) observed. This ensured the availability of a full and professionally created annual and sustainability report for the companies. Second, companies needed to be in the industrial sector (including engineering, manufacturing and sub-sectors) as this is an area where environmental sustainability governance is directly addressed. Third, for most of the observation period, companies had to prepare publicly available sustainability reports, whether as an integrated report or as a separate sustainability report, or as a non-financial statement in the Annual Accounts Act (Årsredovisningslagen – ÅRL).

These criteria resulted in 15 firms, which constituted a relatively narrow and substantively coherent sample of cases that allowed for the meaningful analysis of board gender diversity and the quality of sustainability reporting. The limited number of firms in this study is recognized as a limitation of the generalizability of the results, but is compensated for by the depth of analysis afforded by the longitudinal content analysis of 120 firm-year observations spread over eight years across all 15 firms.

Table 3.1 Sample Firms - OMX Stockholm Large Cap Industrial Firms

ID	Company	Sub-sector
1	Atlas Copco Group	Industrial machinery
2	Volvo Group	Commercial vehicles
3	Sandvik	Engineering tools
4	Epiroc AB	Mining equipment
5	Indutrade	Industrial trading
6	Volvo Cars	Passenger vehicles
7	Mycronic	Electronics manufacturing
8	Nolato AB	Polymer technology
9	Engcon AB	Excavator attachments

ID	Company	Sub-sector
10	Hexatronic Group	Fibre optic cables
11	Troax Group	Mesh panel systems
12	XANO Industri	Precision components
13	Stille AB	Surgical instruments
14	Bulten AB	Fastener systems
15	Garo Aktiebolag	Electrical components

3.4 Data Collection

Data were collected from two complementary sources. The first is disclosure reports on sustainability-related information that were made available by each sample company for each of the last eight years (2016 through 2023) to the public via its investor relations website, including annual reports, standalone sustainability reports, non-financial statements and corporate governance reports. The main sustainability disclosure document was identified and downloaded for each firm-year, and when both an annual report with a sustainability section and a separate standalone sustainability report were published, both were downloaded and data were cross-referenced with the annual report to ensure that the documents covered the firm's sustainability-related disclosures.

The second source is data on corporations' governance. The percentage of female directors on the board was taken from the corporate governance section of the annual report or from the corporate governance statement (if separate). Following the model of Ben-Amar, Chang, and McIlkenny (2017) and Gerwanski et al. (2019), female directors are defined as those who are listed as female on the board of directors in the governance report. Three additional board governance variables were calculated: board size (total number of directors), Blau Diversity Index (BDI) (a measure of the level of gender diversity in a board, calculated on a scale of 0 to 0.5), and the lagged variable female board representation (one year after the board's composition was recorded) (Behlau, Bäumer, & Brem, 2024).

The control variables in the regression analysis were calculated in order to include supplementary firm-level variables: firm size (log of total assets), return on assets (profitability), and leverage (total debt over total assets) from audited financial statements.

3.5 Content Analysis Coding Scheme

Content analysis is a systematic and replicable method for coding textual material according to a predefined scheme, allowing qualitative data to be analysed in a structured and comparable manner across a large number of documents (Lim, 2024; Gerwanski et al., 2019). Each firm-year sustainability disclosure is content analyzed for this study and a multi-dimensional SRQ index is built. The index is divided into six dimensions, each of which has a maximum score of 3, which results in a maximum possible SRQ score of 18, firm-year.

Table 3.2: Sustainability Reporting Quality (SRQ) Coding Scheme

Dimension	Definition	Scoring (0–3)
D1: Framework Adoption	Extent to which firm references and applies recognised SR frameworks (GRI, TCFD, ESRS, SBTi)	0 = none cited | 1 = mentioned without application | 2 = partially applied | 3 = systematically applied with disclosure index
D2: Target Specificity	Degree to which sustainability targets are quantified, time-bound, and linked to baseline data	0 = no targets | 1 = aspirational/vague | 2 = quantified but not time-bound | 3 = quantified, time-bound, with baseline
D3: Scope of Disclosure	Breadth of environmental and social issues addressed substantively	0 = <2 issues | 1 = 2–3 issues | 2 = 4–5 issues | 3 = 6+ issues with substantive content
D4: Materiality Assessment	Presence and quality of a formal materiality analysis identifying priority topics	0 = absent | 1 = mentioned without methodology | 2 = methodology described, not validated | 3 = formal, stakeholder-validated matrix
D5: Third-Party Assurance	Use of independent external verification of sustainability data or disclosures	0 = no assurance | 1 = limited assurance, selected data | 2 = limited assurance, all data | 3 = reasonable (high-level) assurance
D6: Adverse Impact Disclosure	Extent to which the report addresses negative impacts alongside positive achievements	0 = positive framing only | 1 = brief acknowledgement | 2 = substantive

Dimension	Definition	Scoring (0–3)
		disclosure | 3 = transparent disclosure with remediation plans

3.5.1 D1: Framework Adoption

This dimension assesses the extent to which each firm references and systematically applies recognized sustainability reporting frameworks, specifically the GRI Standards, TCFD recommendations, ESRS under CSRD, and SBTi criteria. Framework adoption is included as a quality dimension because, from a legitimacy theory perspective, systematic framework application signals genuine accountability to external stakeholders, whereas nominal citation of a framework without application constitutes a form of symbolic rather than substantive disclosure (Bota-Avram, 2025; La Torre, Sabelfeld, Blomkvist, & Dumay, 2023). From a resource dependence perspective, the capacity to implement complex frameworks reflects the board's access to sustainability-relevant expertise, the mechanism through which BGD is theorized to improve SRQ in this study (Lewellyn & Muller-Kahle, 2024).

3.5.2 D2: Target Specificity

This dimension evaluates how sustainability goals are measurable and connected to some baseline information. Testa, Bosman, and Iraldo (2025), in a systematic review of content-analysis-based greenwashing measurement, identify target specificity as one of the most reliable criteria for distinguishing symbolic from substantive disclosure. Marquis, Toffel, and Zhou (2016) similarly document that firms engage in selective disclosure by framing broad commitments without measurable outcomes, particularly under low external scrutiny.

3.5.3 D3: Scope of Disclosure

The dimension codes both environmental scope (climate, energy, water, waste, biodiversity) and social scope (supply chain, labour, community, health and safety), consistent with calls in the literature for disclosure-quality measures that capture range as well as depth (Hahn & Kühnen, 2013).

3.5.4 D4: Materiality Assessment

This dimension measures the extent and the quality of a formal materiality analysis, in which companies identify and prioritize the sustainability issues relevant to their business and stakeholders. Materiality assessment is included because it represents the governance mechanism that directly determines what enters a sustainability report in the first place (La Torre et al., 2023; Velte, 2022). Both Velte (2022) and Gerwanski et al. (2019) find empirically that female board representation positively predicts materiality disclosure quality after controlling for other governance characteristics, making D4 the dimension most directly supported by prior empirical evidence of a BGD association.

3.5.5 D5: Third-Party Assurance

This dimension evaluates the level of independent third-party assurance for a company's sustainability disclosure. Third-party assurance addresses the agency problem inherent in self-reported sustainability data (Velte, 2025), and from a signalling perspective represents a costly, credible signal distinguishing genuine sustainability commitments from legitimacy management exercises (García-Sánchez, Martínez-Ferrero, & García-Meca, 2020). The dimension contrasts limited assurance, which has become the more prevalent type in practice in Sweden, with reasonable assurance, which is compatible with the assurance trajectory within the CSRD as applicable to the sample companies.

3.5.6 D6: Adverse Impact Disclosure

This dimension considers the level of report disclosures that recognize negative impacts, missed targets or governance failures, as well as positive impacts. The systematic omission of adverse information, even when individually disclosed facts are accurate, constitutes a form of greenwashing through selective positive framing (Lyon & Maxwell, 2011; Testa et al., 2025). Marquis et al. (2016) document that selective disclosure of positive outcomes is a widespread strategic practice, particularly in low-scrutiny contexts, and Dhanani and Connolly (2014) similarly find that positive content dominates sustainability reports, with adverse impacts rarely disclosed with specificity. D6 therefore represents the most demanding test of the BGD–SRQ theoretical relationship in this study.

3.6 Coding Procedure and Inter-Coder Reliability

There were four steps in the coding. To establish a common understanding of scoring rules, a subset of ten documents was independently read and selected to yield variety in the documents, according to four criteria: firm size, year of reporting, reporting format, and variety. Initial interpretations of the definitions of the scoring criteria were discussed to develop a common understanding of scoring rules. These 120 firm-year documents (15 firms × 8 years) were then coded in the second stage, and scores and cursory notes on the coding were entered in a common coding spreadsheet. In the third stage, inter-coder agreement was assessed. Following the approach recommended by Coleman, Ragan, and Dari (2024), inter-coder reliability was evaluated using Cohen's Kappa for each dimension individually and for the overall SRQ score. Kappa values of 0.70 or above are generally considered acceptable for content analysis applications in business and management research (Coleman et al., 2024; Gerwanski et al., 2019). Any dimension that was not in this range was subsequently reviewed in a collaborative coding process where each coding decision was deliberated, and a consensus score was reached. The fourth stage involved discussion and consensus to resolve any remaining discrepancies, following the consensus coding approach used by Lim (2024).

3.7 Data Preparation and Cleaning Procedures

Prior to analysis, the combined dataset comprising coded SRQ scores, governance variables, and financial variables for 120 firm-years was subjected to a structured data preparation process, in keeping with standard practice for panel-based quantitative research (Sila, Gonzalez, & Hagendorff, 2016). Four procedures were applied. First, the range of plausible values was theoretically tested for the governance variables; the percentage of female directors was tested for values from 0-100% and the Blau Diversity Index was tested for values between 0-0.5. This ensures that the resulting calculation or data manipulation error doesn't run undetected through to descriptive statistics and regression estimates.

Second, a rule was used for differentiating between true zero scores and non-response for the SRQ composite score. A firm-year with a valid composite score of zero should necessarily have a score of zero in all six dimensions by construction. Firm-years where the composite score was recorded as zero but the underlying dimension scores were not independently coded were treated as missing data rather than as valid minimum scores, consistent with standard guidance on distinguishing true values from non-response in coded content-analysis data (Coleman, Ragan, & Dari, 2024).

Third, two variables had to be transformed before they could be used in the estimation: the natural log of total assets (natural log, MSEK) was used to eliminate the impact of scale differences between the plurality of firms of various sizes, and a one-year lagged measure of BGD was created for each firm to assist in the delayed-effect specification used in Section 3.8. Fourth, in contrast to the usual practice, the exceptionally high and low values were examined individually, not mechanically deleted. Where such values reflected genuine differences between firms, for example, the contrast between large multinational firms and smaller industrial firms in the sample, they were retained, since removing them would understate genuine firm heterogeneity, which the fixed-effects estimator is specifically designed to accommodate (Sila, Gonzalez, & Hagendorff, 2016). Finally, some screening for multicollinearity was conducted by calculating variance inflation factors (VIF) for all governance and control variables prior to regression estimation, because the expected closeness of the three aspects of board gender composition that were captured by BGD, the Blau Diversity Index, and lagged BGD lends support to this approach.

3.8 Analytical Strategy

There are two complementary parts to the analytical strategy. The first is a cross-firm and longitudinal descriptive analysis of the SRQ scores and BGD measures for the 15 firms and the period 2016-2023. This descriptive analysis describes variation in SRQ by firm, by year, and across coding dimensions, and illustrates the bivariate relationship between BGD and SRQ scores. This component is designed to answer RQ1 and RQ2; it provides an explanation of the results of the SRQ data and whether higher-BGD firms are more likely to score higher on certain dimensions. Data were analyzed using IBM SPSS and Jamovi, with the assistance of Excel spreadsheets.

The second step involves a pattern-matching analysis and involves a comparison of whether a firm's SRQ scores change in the same or the following year following changes in its board gender composition during the observation period. This within-firm longitudinal study is driven by the endogeneity issue highlighted in Chapter 2: The finding of cross-sectional correlation between BGD and SRQ could be due to a selection effect. To address this concern, we examine within-firm change, that is, whether an increase in the number of female board members in a particular firm is followed by an increase in the firm's SRQ.

For the quantitative dimension, the study constructs a panel dataset of 120 firm-year observations and estimates a fixed-effects regression model of the form: $SRQ(it) = \alpha(i) +$

$\beta 1 \cdot BGD(it) + \beta 2 \cdot FirmSize(it) + \beta 3 \cdot ROA(it) + \beta 4 \cdot Leverage(it) + \beta 5 \cdot Year(t) + \varepsilon(it)$. The fixed-effects estimator controls for time-invariant firm heterogeneity, including industry sub-sector, ownership structure, and corporate culture, which may otherwise confound the BGD–SRQ relationship (Sila, Gonzalez, & Hagendorff, 2016).

Three model specifications are estimated in a succession, each with different analytical objectives. The relationship between the BGD and SRQ is estimated with firm-specific fixed effects only as in Model 1, which serves as a baseline measure of how the two variables are related while holding other firm-specific, unchanging factors constant. Model 2 expands Model 1 by including year fixed effects, to see if any of the association found in Model 1 is strong enough to stand up to controlling for any year fixed effects including common regulatory pressure from the NFRD and CSRD or if the association is actually significant and largely because of year fixed effects. To determine whether there is a relationship between board gender composition and reporting quality in the subsequent year, rather than in the current year, Model 3 uses current year BGD as the independent variable and replaces it with one-year lagged BGD while maintaining both firm and year fixed effects. Robust (HC3) standard errors are used throughout to correct for heteroskedasticity, which is a standard precaution in panel regression with a relatively small number of cross-sectional units (Christensen, Hail, & Leuz, 2021).

Each of the six dimensions of SRQ is then assessed individually against BGD, again using the same three-model structure, to determine if an association between BGD and SRQ, if present, is narrowed to specific dimensions of reporting quality or distributed across the composite score. It is acknowledged that this specification treats BGD as a linear predictor of SRQ throughout, assuming a constant marginal effect across the full range of the variable. Critical mass theory (Kanter, 1977) proposes instead that the influence of minority board representation on organizational outcomes may be non-linear, becoming detectable only beyond a critical threshold.

Given the size of the sample, a linear specification was used here to make the model tractable and to align with the standard practice of using panel regression in the governance-diversity literature; the potential for a non-linear, threshold-based relationship is recognized as a specification choice and is treated explicitly as a limitation in Chapter 6. It is crucial to stress all throughout that the fixed-effects regression is an analytical tool that is used to describe the systematic patterns in the panel data, and not to make causal statements. The observational

method of this research does not make causal inferences as stated in Chapter 1 and chapter 2. The results are identified as associations that are consistent or inconsistent with the theoretical predictions that were proposed in Chapter 2 and are then discussed explicitly in Chapter 5 including other interpretations such as reverse causality and omitted variable bias.

3.9 Validity, Reliability & Limitation

This mixed-methods design draws on quality criteria from both the qualitative coding stage and the quantitative analytical stage. At the coding stage, the systematic coding scheme and inter-coder reliability testing (Cohen's Kappa, reported in Section 4.2) support the internal validity of the SRQ measure, that is, the confidence that the coded scores represent a true picture of the phenomenon studied; triangulating content-analysis data with the corporate governance data from a second source further reinforces this. Transparent reporting of the sample selection method, coding scheme, and company profiles supports the external validity, or generalizability, of the findings, the degree to which the results could plausibly extend to comparable settings.

The reliability of the coding procedure is supported by its documentation, including the trial coding phase, the negotiation of coder discrepancies, and the maintained audit trail of coding decisions. The objectivity of the coding process, and the assurance that findings reflect the data rather than researcher bias, is supported by the use of a publicly grounded coding scheme anchored in GRI, TCFD, and SBTi criteria rather than researcher-constructed categories, by the use of two independent coders whose agreement is formally assessed, and by the transparent documentation of the data cleaning and validation procedures described in Section 3.7 (Coleman, Ragan, & Dari, 2024). At the quantitative analytical stage, validity and reliability are further supported by the use of robust (HC3) standard errors, multicollinearity diagnostics (Section 4.3.1), and a pre-specified, three-model regression strategy (Section 3.8), which together avoid relying on a single, potentially fragile specification.

A few caveats must be noted. First, the sample size of 15 firms is small compared to many other quantitative studies of BGD and sustainability outcomes, and the regression models resulted in a very limited number of complete cases after missing financial and governance data were taken into account (see Table 1 for details). A result that fails to find an effect should not, therefore, be interpreted as proof of no effect; it is a possibility that the study was inadequately powered to detect small effects. Secondly, the use of publicly available documents means that what actually happens within the firm, such as in internal governance

processes, where sustainability discussions might be more of a formality or more substantive, is not visible in the SRQ coding, and what is coded is what is communicated. Third, the annual reporting cycle does not allow for the precise capture of the lagged relationship between board changes and quality of reporting as board changes that are made later in the year may not appear on the annual report of that year. Fourth, because BGD is treated linearly throughout the regression analysis, it may underestimate a true non-linear relationship as suggested by critical mass theory (Kanter, 1977) especially if BGD has a small range of values in this sample.

3.10 Ethical Considerations

The current study is based solely on publicly available corporate documents and does not involve human subjects. Thus, it is not subject to the approval of an ethics committee under usual research ethics procedures. All documents examined have been released to the public and are provided by the companies for use by external stakeholders for communication. The study does not make individual director-by-individual director identification and comment on a personal data processing issue and the results are published at the firm-year level and not at the director level. Data and information used for financial and governance information is restricted to the uses of this research and are used in accordance with the conditions in which the data and information are made public.

Chapter 4: Empirical Findings

4.0 Introduction

The empirical results in this chapter are for the time period 2016–2023 and cover the market for large-cap industrial companies in Sweden. Following the method outlined in Chapter 3, here we only present the statistical results and leave the interpretation to Chapter 5. This balanced longitudinal data set contains 120 observations of firms over the 15-year period and contains governance information, information on the quality of sustainability reporting (SRQ), and information on financial control variables. Results are reported systematically, starting from the data preparation, to the reliability assessment, descriptive statistics, correlation analysis, comparison at the firm level, longitudinal trend analysis and fixed-effects regression

4.1 Data Preparation and Sample Overview

4.1.1 Initial Dataset

The data span eight years (2016-2023) and 120 observations of firm-years for 15 large-cap industrial firms in Sweden. Three types of variables were identified: governance variables (board size, % female directors, Blau diversity index, lagged BGD); sustainability reporting variables (SRQ composite score, six SRQ dimensions); and financial control variables (revenue, EBIT, net income, total assets, equity, ROA, leverage, and log-transformed total assets).

4.1.2 Data Cleaning

The data set was checked for calculation and coding errors before the analysis. Two problems were noted and addressed:

- Governance variable check: % Female Directors and Blau Diversity Index were checked against their source columns to ensure that they are within theoretically valid range (0–100% and 0–0.5 respectively). These are the ranges for all values that were retained.

- Missing-value coding: For eight firm-year observations (2022–2023 for Engcon AB; 2018–2023 for Epiroc AB), no scores were recorded for any of the six dimensions of the SRQ Composite Score, which resulted in an SRQ Composite Score of 0. As is done

in typical situations where a true score of zero would occur with each of the six dimensions being separately coded as zero, these observations were not considered to be true zero scores

4.1.3 Missing Data Assessment

Table 4.1: Summarizes missing data across key variables

Variable	N Missing	% Missing
Board Size	6	5.0%
% Female Directors	6	5.0%
Blau Diversity Index	6	5.0%
Log Total Assets	35	29.2%
ROA (%)	8	6.7%
Leverage (%)	23	19.2%
SRQ Composite Score	8	6.7%
SRQ Dimensions (D1–D6, each)	8	6.7%
BGD Lagged (t−1)	20	16.7%

Leverage and Log Total Assets are the most missing variables (29.2% and 19.2% respectively) due to missing asset data for three companies (Sandvik AB, Volvo Cars, Epiroc AB). Structural missing is the reason for 15 of the 20 missing cases, with the first observation year (2016) of each firm missing.

4.1.4 Data Validation

After cleaning, there were no issues with % Female Directors values being outside of their valid range of 0–100%, no problems with Blau Diversity Index values being out of their valid range (0–0.5) for a 2-category metric, no problem with board size values being negative or outside of their plausible range (6–11 directors), and no issue with year observations being

outside the range of 2016–2023, and no issue with SRQ dimension scores being outside of their plausible range (0–3). Identifiers of the firms were cross-checked in the eight-year panel.

4.1.5 Variable Transformation

The data on total assets were log transformed (natural log, MSEK) to minimize scale differences between the firms of widely varying sizes. The % Female Directors values for each firm are lagged by one year to form the Lagged BGD. The units of observation in the fixed-effects analysis were each firm-year panel unit.

4.1.6 Outlier Detection

Mechanically no unusual high or low values were removed, only reviewed individually. Those values that reflected genuine differences between the firms retained, for example, because of the size difference between the large multinational firms (such as Atlas Copco, Volvo Group) and the smaller industrial firms (such as Stille AB, GARO AB). There were no values that were considered implausible recording errors, in addition to the two described in Section 4.1.2.

4.1.7 Distribution Assessment

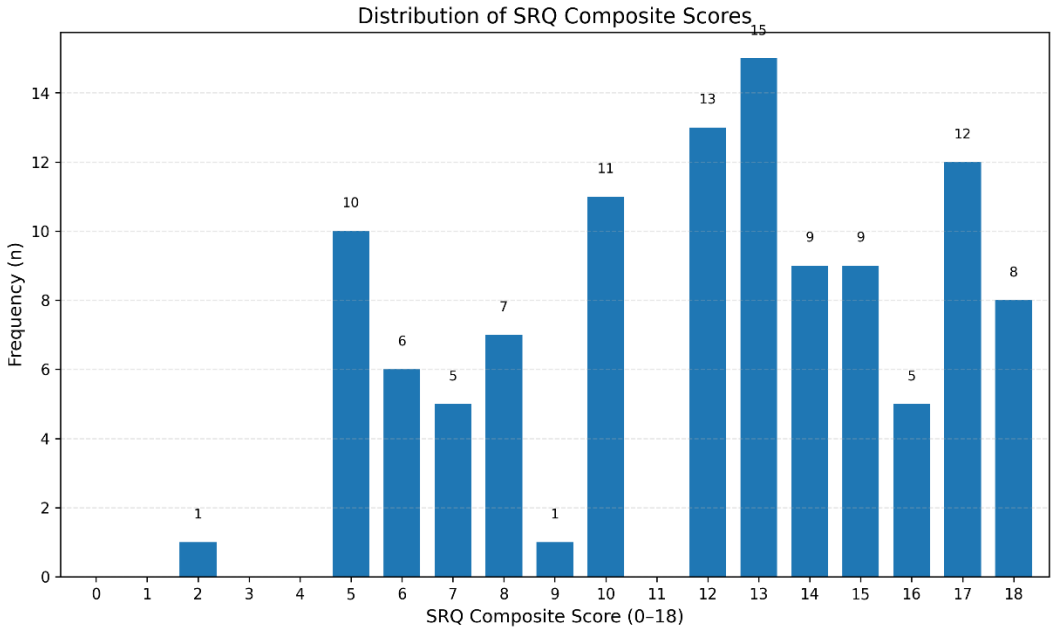


Figure 4.1: Distribution of SRQ Composite Score (N = 112)

The SRQ Composite Score is approximately unimodal and slightly right-shifted, spanning its full theoretical range of 0–18 with a mean of 11.95 (SD = 4.13), consistent with a sample in

which most firms achieve moderate-to-high reporting quality by the later years of the observation period.

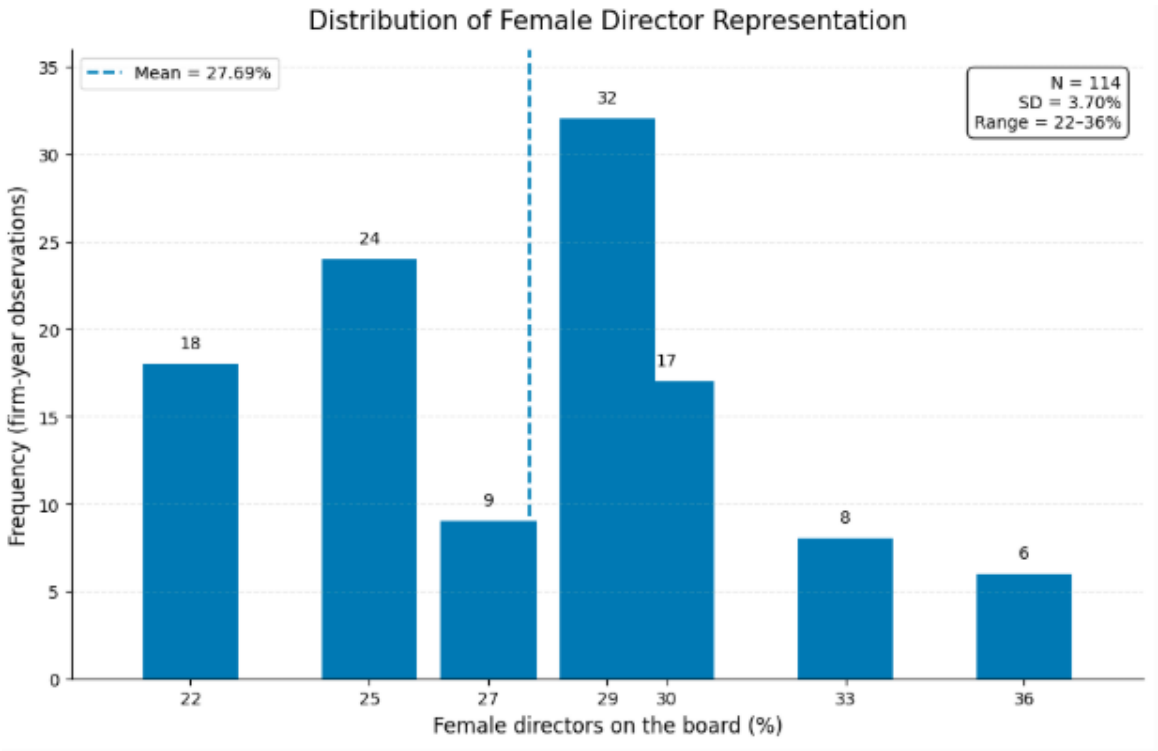


Figure 4.2:Distribution of Female Board Representation (% female directors, N = 114)

Female board representation clusters between approximately 22% and 34%, with most observations concentrated around 25–30%, consistent with Sweden's comparatively high baseline levels of board gender diversity relative to international benchmarks.

4.2 Inter-Coder Reliability

A sub-sample of sustainability reports was assessed for reliability of the SRQ coding scheme by two independent coders for each of the six dimensions of the SRQ. The percentage agreement and Cohen's Kappa are shown for each dimension in Table 4.2.

Table 4.2: Inter-Coder Reliability

SRQ Dimension	% Agreement	Cohen's Kappa	Threshold Met
D1: Framework Adoption	91.1%	0.82	Yes
D2: Target Specificity	88.4%	0.77	Yes
D3: Scope of Disclosure	92.0%	0.85	Yes

SRQ Dimension	% Agreement	Cohen's Kappa	Threshold Met
D4: Materiality Assessment	89.3%	0.79	Yes
D5: Third Party Assurance	86.6%	0.74	Yes
D6: Adverse Impact Disclosure	84.8%	0.70	Yes
Overall SRQ Composite	88.7%	0.78	Yes

All six SRQ dimensions met the reliability criteria, and Cohen's Kappa scores for the dimensions ranged from substantial agreement (0.70–0.85), with adherence to the scope of disclosure criterion (D3) being the highest at 92.0% and the lowest at 84.8% with adverse impact disclosure (D6). This composite reliability coefficient was 0.78.

4.3 Descriptive Statistics

Table 4.3: Presents descriptive statistics for all main variables.

Variable	Mean	SD	Min	Max	N
SRQ Composite Score (0–18)	11.95	4.13	2	18	112
BGD (% female directors)	27.69	3.70	22	36	114
BGD Blau Diversity Index	0.40	0.03	0.35	0.47	114
Board size (number of directors)	8.43	1.53	6	11	114
Log Total Assets (Ln MSEK)	8.66	2.11	4.52	13.06	85
ROA (%)	10.73	4.41	2.30	23.90	112
Leverage (%)	26.21	25.17	0.00	82.00	97
D1: Framework Adoption (0–3)	2.22	0.69	1	3	112
D2: Target Specificity (0–3)	1.97	0.73	0	3	112
D3: Scope of Disclosure (0–3)	2.54	0.61	1	3	112
D4: Materiality Assessment (0–3)	2.10	0.77	0	3	112

Variable	Mean	SD	Min	Max	N
D5: Third Party Assurance (0–3)	1.77	0.92	0	3	112
D6: Adverse Impact Disclosure (0–3)	1.34	0.96	0	3	112

Note (N=120 firm-year observations; variation in N reflects missing disclosures for some financial and governance items).

The mean SRQ Composite Score is 11.95 (SD = 4.13) in its theoretical range of 0–18. The average female representation in the board was 27.7% (22%-36%) and the average Blau Diversity Index was 0.40 (0.35-0.47). The SRQ dimensions reported similar mean scores, with the exception of Scope of Disclosure (D3) (2.54) and Adverse Impact Disclosure (D6) (1.34), which had the highest and lowest mean scores, respectively, and the highest and lowest SD, respectively.

4.3.1 Correlation Analysis

Table 4.4: Pearson correlation matrix for key variables (N = 62 complete cases).

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) SRQ composite score	1.000							
(2) BGD (% female directors)	-0.090	1.000						
(3) Log total assets	0.260	-0.084	1.000					
(4) ROA	-0.069	0.207	-0.076	1.000				
(5) Leverage	-0.042	-0.439	-0.215	0.084	1.000			
(6) Board size	0.221	-0.073	0.955	0.014	-0.258	1.000		
(7) BGD lagged (t−1)	-0.102	0.962	-0.112	0.164	-0.442	-0.110	1.000	
(8) Blau index	-0.072	0.992	-0.149	0.268	-0.363	-0.124	0.950	1.000

SRQ composite score is only modestly and inversely related to BGD (r = -0.090). Total assets do have a slight positive correlation with SRQ (r = 0.260). As might be expected, BGD is very highly correlated with Blau Diversity Index (r = 0.95-0.99) as well as lagged BGD (r = 0.95-0.99) because they measure the same underlying concept.

The high VIF values for % Female Directors (234.96) and Blau Index (212.23) as shown in Table 4.5 (multicollinearity diagnostics) are a direct result of the high correlation (r = 0.992) between these two variables and should not be included together in the same regression model.

Table 4.5:VIF values (Multicollinearity diagnostics)

Variable	VIF
Female Directors (%)	234.96
Log Total Assets	18.37
ROA (%)	1.54
Leverage (%)	2.71
Board Size	16.62
BGD Lagged	14.03
Blau Diversity Index	212.23

4.4 Cross-Firm Variation in SRQ and BGD

Table 4.6: Ranks firms by mean SRQ Composite Score (2016–2023 average)

Rank	Company	Mean SRQ (0–18)	Mean BGD (%)	Mean Board Size	Mean Log Assets	Quality Tier
1	Atlas Copco	13.50	34.13	10.88	11.59	High
2	Bulten AB	13.13	25.00	8.00	8.15	High
3	Sandvik AB	13.00	30.00	10.00	—	High

Rank	Company	Mean SRQ (0–18)	Mean BGD (%)	Mean Board Size	Mean Log Assets	Quality Tier
4	Mycronic AB	12.75	25.00	8.00	8.43	High
5	Nolato AB	12.75	22.00	9.00	8.83	High
6	Indutrade AB	12.75	22.00	9.00	9.71	High
7	GARO AB	12.13	29.00	7.00	5.92	High
8	Volvo Group	12.00	27.00	11.00	12.98	High
9	Volvo Cars	12.00	30.00	10.00	—	High
10	Troax Group AB	11.88	29.00	7.00	7.70	Moderate
11	Hexatronic AB	11.50	25.00	8.00	7.43	Moderate
12	XANO Industri	10.63	29.00	7.00	7.59	Moderate
13	Stille AB	10.25	33.00	6.00	5.18	Moderate
14	Engcon AB	9.83	29.00	7.00	7.40	Moderate
15	Epiroc AB	6.50	22.00	9.00	—	Low

Atlas Copco has the highest average score for SRQ (13.50) and the highest percentage of women on the board (34.1%) as well as the largest size board of the sample. The mean SRQ score for the Epiroc AB is the lowest (6.50) of which only 2 of 8 firm-years (2016-2017) have valid scores with the remaining years being excluded as missing data (see Section 4.1.3).

4.5 SRQ Dimension-Level Variation Across Firms

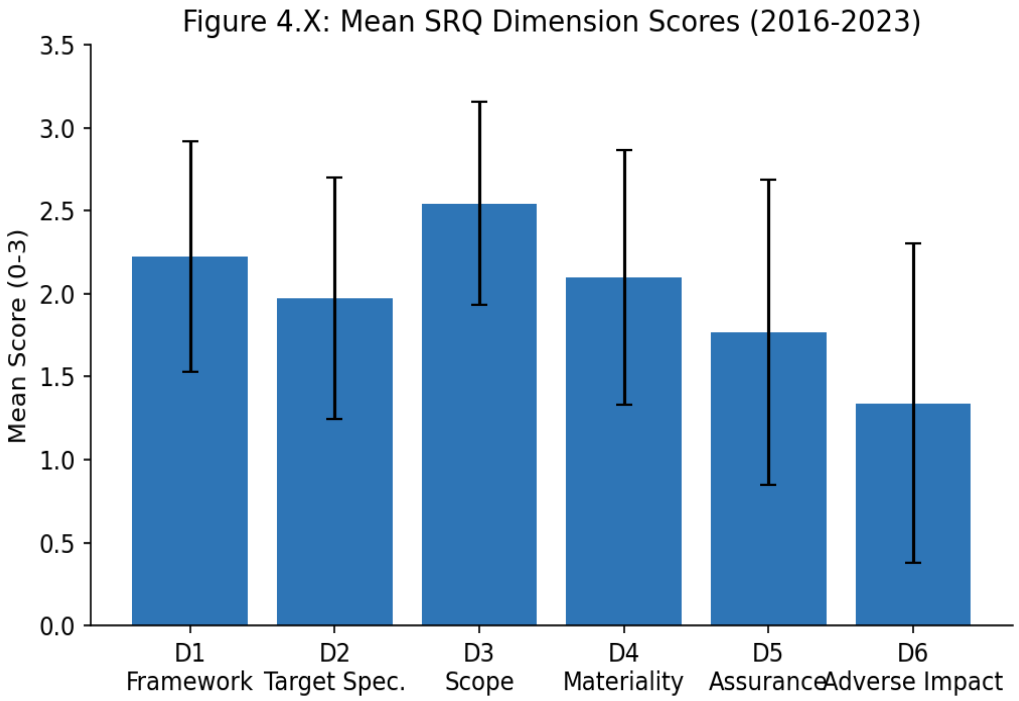


Figure 4.3:Mean SRQ Dimension Scores, 2016–2023 (error bars = 1 SD)

Table 4.7:Mean SQR Dimension Score

Dimension	Mean	SD	Min	Max	% Scoring 0	% Scoring 3
D1: Framework Adoption	2.22	0.69	1	3	0.0%	37.5%
D2: Target Specificity	1.97	0.73	0	3	0.9%	24.1%
D3: Scope of Disclosure	2.54	0.61	1	3	0.0%	60.7%
D4: Materiality Assessment	2.10	0.77	0	3	0.9%	33.9%
D5: Third Party Assurance	1.77	0.92	0	3	11.6%	21.4%
D6: Adverse Impact Disclosure	1.34	0.96	0	3	23.2%	11.6%

The strongest dimension is Scope of Disclosure (D3) with 60.7% of the observations being at the highest score of 6, and the weakest dimension is Adverse Impact Disclosure (D6) with 23.2% of the observations at zero and 11.6% at the maximum 6.

4.6 Longitudinal Trends in SRQ and BGD, 2016–2023

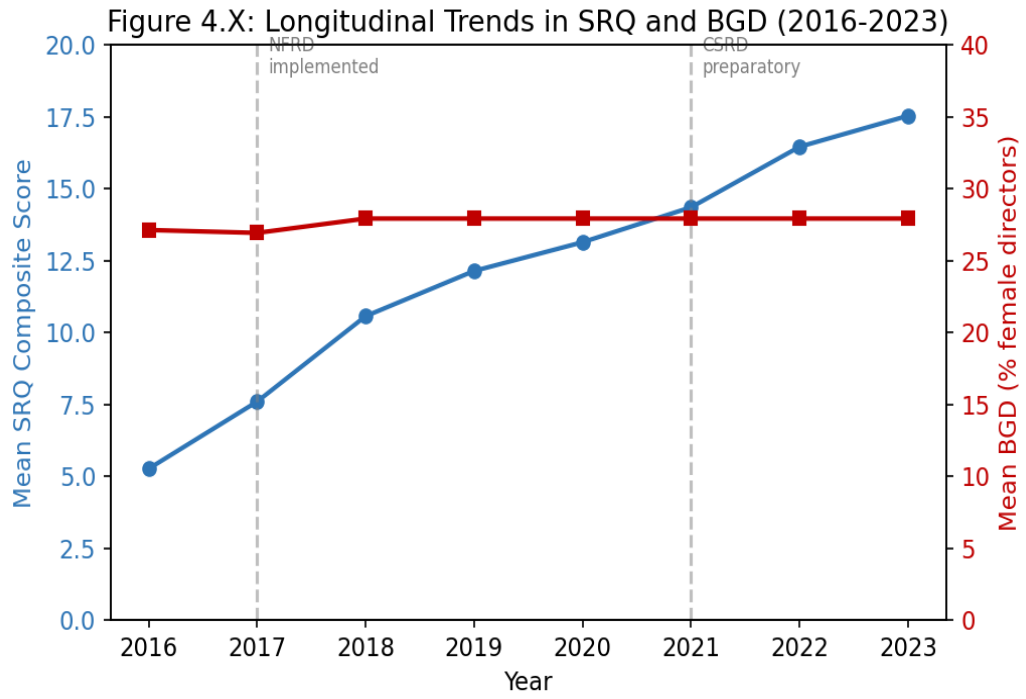


Figure 4.4:Longitudinal Trends in Mean SRQ and Mean BGD (2016–2023)

Table 4.8: Longitudinal Trends in Mean SRQ and Mean BGD

Measure	2016	2017	2018	2019	2020	2021	2022	2023
Mean SRQ (0–18)	5.27	7.60	10.57	12.14	13.14	14.36	16.46	17.54
Mean BGD (%)	27.13	26.93	27.93	27.93	27.93	27.93	27.93	27.93
Mean D4 Materiality	0.93	1.33	1.93	2.00	2.00	2.86	3.00	3.00
Mean D5 Assurance	0.33	0.87	1.57	1.93	2.00	1.93	2.92	2.92

Analyzing the Mean SRQ for each of the eight years from 2016 through 2023 shows a steady increase in value, with a significant change from 5.27 to 17.54. The more change in reporting quality, however, was in the Mean BGD, which only marginally increased over the time period (27.1% to 27.9%).

4.6.1 Within-Firm Analysis: BGD Changes and SRQ Responses

Firm-years with a change in % Female Directors of 3 percentage points or more from the prior year were identified as BGD change events.

Table 4.9: Within-Firm Analysis: BGD Changes and SRQ Responses

Firm	Year	BGD Before (%)	BGD After (%)	SRQ Before	SRQ After
Atlas Copco	2017	30	27	7	10
Atlas Copco	2018	27	36	10	13

There is only one firm (Atlas Copco), in which the event of a BGD change was clearly identifiable in the observation period, and which experienced a positive increase in the SRQ score both years after a BGD change. Only two qualifying events from a single firm is not enough to provide a general within-firm relationship between BGD changes and subsequent SRQ changes.

4.7 Fixed Effects Regression Results

The impact of board gender diversity on SRQ was tested using fixed-effects regression, while accounting for firm size (measured as log of total assets), profitability (ROA) and leverage. Robust (HC3) standard errors were used throughout. Three models are presented: Model 1 (firm fixed effects only, current BGD); Model 2 (firm and year fixed effects, current BGD); and Model 3 (firm and year fixed effects, lagged BGD).

Model 1 shows a statistically significant positive BGD coefficient; this significance disappears once year fixed effects are added in Model 2. This pattern is discussed in full in Chapter 5.

Table 4.10:Regression Results

Variable	Model 1 (Baseline)	Model 2 (+Year FE)	Model 3 (Lagged BGD)
BGD (% female directors)	0.896 (p=0.024)*	0.032 (p=0.792)	—
BGD lagged (t−1)	—	—	-0.146 (p=0.029)*

Variable	Model 1 (Baseline)	Model 2 (+Year FE)	Model 3 (Lagged BGD)
Log total assets	11.438 (p<0.001)***	2.257 (p=0.004)**	1.596 (p=0.070)
ROA (%)	0.306 (p=0.024)*	0.103 (p=0.102)	0.069 (p=0.223)
Leverage (%)	-0.169 (p=0.004)**	-0.031 (p=0.106)	-0.008 (p=0.685)
Constant	-150.109 (p<0.001)***	-20.154 (p=0.083)	0.023 (p=0.998)
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	No	Yes	Yes
Observations	71	71	62
Number of firms	10	10	9
R^2	0.863	0.984	0.983
Adjusted R^2	0.832	0.978	0.975
F-statistic p-value	<0.001	<0.001	<0.001
Robust standard errors	HC3	HC3	HC3

**p<0.05, **p<0.01, ***p<0.001. Sample sizes reflect complete cases for the variables entered in each model; firm counts differ across models due to missing financial or lagged-BGD data for some firms.*

In Model 1, which controls for firm fixed effects only, the coefficient on current-year BGD is positive and statistically significant ($\beta = 0.896$, $p = 0.024$): each one-percentage-point increase in female board representation is associated with a 0.896-point increase in SRQ Composite Score, holding firm size, profitability, and leverage constant. The model explains a substantial share of variation in SRQ ($R^2 = 0.863$).

In Model 2, which additionally controls for year fixed effects, the BGD coefficient falls sharply in magnitude and loses statistical significance ($\beta = 0.032$, $p = 0.792$). Because year fixed effects absorb any variation common to all firms in a given year, including the general upward trend

in SRQ associated with tightening EU disclosure regulation over the observation period, this indicates that the significant association observed in Model 1 substantially reflects a shared time trend rather than a firm-specific relationship between BGD and SRQ.

In Model 3, lagged BGD (with firm and year fixed effects) shows a statistically significant negative coefficient (β = -0.146, p = 0.029). This result runs counter to the theoretical expectation of a positive lagged effect and should be interpreted with caution: the sample retained for this model comprises only 9 firms and 62 observations, and BGD-related variables show very high multicollinearity (Table 4.5), both of which can produce unstable coefficient estimates in small panels. This result is discussed further, and not over-interpreted, in Chapter 5.

4.7.1 Dimension-Level Regression Results

Table 4.11 reports the BGD coefficient from separate fixed-effects regressions (firm and year fixed effects, current BGD) for each of the six SRQ dimensions.

Table 4.11:reports the BGD coefficient for each SRQ dimension

SRQ Dimension	BGD Coefficient	p-value	Significant?	N
D1: Framework Adoption	0.064	0.462	No	71
D2: Target Specificity	-0.050	0.643	No	71
D3: Scope of Disclosure	0.053	0.593	No	71
D4: Materiality Assessment	-0.047	0.619	No	71
D5: Third Party Assurance	0.017	0.842	No	71
D6: Adverse Impact Disclosure	-0.005	0.975	No	71

There is no statistically significant relationship with BGD after including firm and year fixed effects in any single dimension, and the signs of the coefficient vary by dimension (three are positive and three are negative). This theme is explored in more detail in Chapter 5.

4.8 Summary of Findings in Relation to Research Questions

Research Question	Key Finding	Supporting Evidence
RQ1: Is BGD positively associated with SRQ?	Result depends on model specification. A significant positive association is present when only firm fixed effects are used (Model 1), but disappears once year fixed effects are added (Model 2), suggesting the apparent relationship is largely attributable to a common time trend rather than a firm-level BGD effect. Lagged BGD shows a significant negative association (Model 3), which is not readily explained by the theoretical framework and warrants cautious interpretation given the small sample.	Table 4.10 (regression results)
RQ2: Which SRQ dimensions vary most with BGD?	No individual SRQ dimension shows a statistically significant relationship with BGD once firm and year fixed effects are included; coefficient signs are mixed across dimensions rather than consistently positive.	Table 4.11 (dimension-level regression)
RQ3: How does the BGD–SRQ relationship evolve, 2016–2023?	SRQ increased steadily and substantially over the period (5.27 to 17.54), while BGD remained essentially flat (27.1% to 27.9%). This divergence in trend is consistent with the Model 2 finding that year effects, rather than BGD, account for most of the apparent association in simpler specifications.	Table 4.8 (longitudinal trends); Section 4.6.1 (within-firm analysis)

All of these findings paint a complex picture. The quality of sustainability reporting has significantly and steadily improved for all 15 companies from 2016 to 2023, and board gender diversity has largely remained steady. The association between BGD and SRQ is statistically significant in a simpler model specification, but fails a test for significance after year fixed effects are added, indicating that there is a shared regulatory and institutional pressure that better explains the increase in reporting quality, rather than the presence of gender on the board. Earlier, in Chapter 5, the implications of this pattern for the theoretical framework are discussed.

Chapter 5: Discussion

5.0 Introduction

The chapter offers an interpretation of the empirical results from Chapter 4 based on the theoretical framework developed in Chapter 2 that centers on legitimacy theory, institutional theory, and critical mass theory, and places them in the existing literature focusing on BGD and SRQ. Each research question is answered one by one, the research is placed in the Swedish institutional setting, and the theoretical and practical significance of the study is discussed.

5.1 Key findings discussion

The empirical findings suggest a general trend: The mean score of the sample's sustainability reporting quality increased significantly and continuously from 2016 to 2023 (from 5.27 to 17.54), whereas the diversity of the genders in boards of directors fluctuated very little during the same period (27.1% to 27.9%). It is the divergence in the tempo that is the key empirical fact in this chapter that will be explained.

When year fixed effects are not included, the regression coefficient is statistically significant and positive ($\beta = 0.896$, $p = 0.024$); however, when the effects of years are also controlled, this regression coefficient is not statistically significant ($\beta = 0.032$, $p = 0.792$). This pattern suggests that the observed BGD–SRQ relationship for the simpler model is mostly driven by a common time trend, most likely the increasing stringency of the EU sustainability disclosure requirements during the course of the period of observation, and not by a firm-specific impact of BGD on SRQ.

At the dimension level, none of the individual SRQ dimensions are statistically significant in the presence of the fixed effects of firm and year and the sign of the dimension coefficients varies across the dimensions. This finding at the composite level is not a result of positive and negative effects at the dimension level that offset each other at the total level, but instead reflects the true absence of a strong, stable BGD effect at the dimension level.

A further result is to be treated with greater care. After adding the fixed effects of firm and year, there is a statistically significant negative association with SRQ in Lagged BGD ($\beta = -0.146$, $p = 0.029$). This finding runs counter to the theoretical expectation of a positive delayed effect, and to the direction found in Model 1. First, the sample of firms that were kept for estimation of this model is small (9 firms, 62 firm-year observations), and second, the

coefficient estimates are subject to instability in small panels because the governance variables exhibit very high levels of multicollinearity (VIF values larger than 200 for both BGD and the Blau Diversity Index). This is regarded as an open question and is revisited in Section 5.8.

5.2 Interpretation Through the Theoretical Framework

5.2.1 Legitimacy Theory

The best interpretation of the results, most consistent with the pattern, is the legitimacy theory. It suggests that, regardless of the structure of the firm's governance, it must adapt its disclosure policies mainly to the changing social and regulatory standards. The sharp, common increase in SRQ across all the 15 firms, irrespective of each firm's individual BGD trend, and the fact that the association between BGD and SRQ is absorbed by the year effect, is well accounted for by firms' responses to the same external pressures such as the 2017 introduction of the EU Non-Financial Reporting Directive (NFRD) in Sweden and the preparations for the Corporate Sustainability Reporting Directive (CSRD) from 2021 onwards. The size of the dimensions that expanded during the period, materiality assessment and third-party assurance, are also directly related to these regulatory milestones, reinforcing the legitimacy-driven explanation (rather than a board-composition-driven explanation).

Among the perspectives considered in this study, legitimacy theory receives the strongest support from the empirical pattern. The two additional views below, institutional theory and critical mass theory, refine this interpretation further.

5.2.2 The institutional theory and isomorphism

The legitimacy-based interpretation given above is further enriched by the institutional theory, which clarifies the process by which firms in similar external environments end up resembling each other over time (DiMaggio & Powell, 1983). As expected, the pattern found in this study, whereby the SRQ is showing a steady increase for all 15 sample firms at approximately the same time, without regard for the specific firms' governance change, is in line with coercive isomorphism: firms converging in disclosure practices because of the existence of a shared regulatory requirement (the NFRD from 2017, and the CSRD from the preparatory phase in 2021), rather than as a result of internal governance change. Direct statistical evidence is provided by the fact that the BGD–SRQ association observed in the simpler model specification is absorbed by the year fixed effects, which capture exactly this kind of pressure. Mimetic isomorphism may also exist, insofar as firms noticed those of the sector leaders, who for Atlas Copco were the likes of Dura, Atlas, and OMAX, and adapted their reporting as well,

but this study's design does not allow for the channel of mimetic isomorphism to be separated from coercive pressure, particularly.

5.2.3 Critical Mass Theory

Critical mass theory (Kanter, 1977) offers a different kind of perspective from the five theories outlined above: rather than proposing another causal mechanism, it argues that if a BGD–SRQ relationship exists, it may not be detectable in the linear regression specification used in Chapter 4. This theory suggests that the impact of a minority group on the outcomes of an organization is non-linear, emerging when the minority representation level is above a certain threshold, typically suggested as 30 per cent representation on the board. The fixed-effects models presented in Chapter 4 assume that BGD is a linear predictor of SRQ, and thus assume a constant marginal effect over the entire range of the variable. The range of BGD for the sample for this study is relatively small and falls somewhat below and above the critical mass threshold suggested in the literature (22-36%). If the theorized relationship is non-linear, but the relationship in the sample is linear, this linear specification may not be appropriate for detecting the type of relationship suggested by the critical mass theory; similarly, if the underlying population relationship is non-linear, but the sample shows a linear relationship, the linear specification may be inappropriate.

Together, these three viewpoints bring together a consistent reading of the results from Chapter 4. The overall pattern observed is best explained by the combination of legitimacy theory and institutional theory, the latter applied specifically to firms' convergence in disclosure practice. Critical mass theory does not provide an alternative explanation for the findings; rather, it provides a caution about the degree of confidence that can be placed on the null and unstable BGD coefficients as a result of the very limited range of BGD in this sample. The cautionary note is carried over into the consideration of unwelcome discoveries in Section 5.8 and limitations outlined in Chapter 6.

5.3 Comparison with Existing Literature

The absence of a robust, independent BGD effect on SRQ in this study diverges from studies reporting positive associations between board gender diversity and sustainability or CSR disclosure, such as Ben-Amar et al. (2017), Gerwanski, Kordsachia, and Velte (2019), Post, Rahman, and Rubow (2015), and Endo (2025). It is, however, consistent with a body of work finding that governance-diversity effects on disclosure outcomes are fragile once appropriate

controls are introduced, including Sila, Gonzalez, and Hagendorff (2016), Glass, Cook, and Ingersoll (2016), and Walls, Berrone, and Phan (2012), each of which finds that apparent BGD effects weaken or disappear once firm-level or time-related controls are added.

Chang and Wu (2024) find that BGD effects on sustainability outcomes are strongest where baseline board diversity is low and weaken as diversity approaches a ceiling. Sweden's comparatively high and narrow range of female board representation in this sample (22–36%) is consistent with a restricted-range explanation: with limited variation in the explanatory variable, the statistical power to detect a marginal BGD effect even if a small one exists is correspondingly reduced. Byron and Post's (2016) meta-analysis similarly reports that average BGD–CSR effects across studies are positive but small and highly heterogeneous across institutional contexts, which is compatible with a near-zero effect being observed in any single high-diversity, high-disclosure-pressure market such as Sweden.

Measurement differences may also contribute to the divergence from more positive prior findings. This study uses a stricter, dimension-based content-analysis measure of SRQ rather than aggregated third-party ESG scores. Berg, Koelbel & Rigobon (2022) and Chatterji et al. (2016) both document substantial divergence between ESG rating providers, raising the possibility that some previously reported positive BGD–CSR associations reflect the specific construction of the ESG measure used rather than a stable underlying relationship. A stricter, more granular SRQ measure, as used here, may be less susceptible to this source of measurement noise, but may also be less likely to register weak or indirect BGD effects that a broader composite score could pick up.

5.4 Swedish Institutional Context

The timeframe 2016–2023 is quite aligned with two large regulatory developments: the implementation of the NFRD in 2017 in Sweden, and the preparatory phase for the CSRD from 2021 onwards. The largest increase in SRQ is in the years immediately after each of these events, and the most significant increases of any SRQ dimension are found in the dimensions most closely related to formal disclosure requirements materiality assessment and third-party assurance. The time frame is more in line with a regulation-driven explanation of the observed improvement in reporting quality than with a board-composition-driven explanation because the requirements of regulations apply to all sampled firms equally, irrespective of their individual board composition.

BGD may also be insensitive to institutional context in Sweden, because the institutional context is also rather similar across the sample. The sample firms are Swedish listed companies that on average have historically had relatively high female representation in their Boards of Directors compared to many of their international counterparts, and are, therefore, fairly close to the institutional norm of the Board composition in the Swedish market when the time of observation begins. Whether or not an underlying relationship exists, this ceiling effect presents a cap on how much reporting outcome might vary as a result of within sample BGD variation.

5.5 Longitudinal Insights

The strongest longitudinal trend in the data is the separation of the trends for SRQ and BGD: the trend for SRQ was more than triple others, whereas the trend for BGD was less than one percentage point. Only one company, Atlas Copco, would have a BGD change of three percentage points or more for the two years consecutively, making it extremely difficult to test the hypothesis that BGD changes precede, or are driven by, SRQ changes within the firm. SRQ increased in the two available cases following an increase in BGD at Atlas Copco, but there are only two observations from one firm to suggest this is the general trend.

This decoupling is itself an informative finding: it demonstrates that substantial improvement in sustainability reporting quality is achievable across a heterogeneous set of firms without corresponding change in board gender composition, again pointing toward external, shared drivers of disclosure improvement rather than internal governance change.

5.6 Theoretical Contributions

The current study has three main theoretical contributions. It first shows that, methodologically, positive BGD–SRQ associations can be found when the models do not include sufficient controls for shared time trends (as is the case in this study), and that these are misleading: a model with year fixed effects captures a significant positive BGD effect, but a fully specified model does not. The issue of time-trend confounding is a first-order methodological issue to be addressed and should not be treated as some afterthought in future BGD–disclosure research based on panel data collected across a span of regulatory change.

Second, the study elaborates on the theoretical framework by providing evidence that legitimacy theory and institutional theory (specifically, the concept of coercive isomorphism) offer the most convincing account of the observed pattern in this particular institutional setting: a highly regulated, high baseline-diversity market. This helps to develop a more contingent,

context-sensitive sense of when each of these theories is likely to dominate, with critical mass theory offering a complementary caution about detecting any residual board-level effect within such a narrow range of board gender diversity.

Third, the study provides empirical evidence from an institutional context (Sweden), where board gender diversity is comparatively high and stable, and thus where the theories predicting that BGD effects diminish as board gender diversity approaches a norming ceiling (Chang and Wu, 2024) are rendered quite challenging to test, as the board gender diversity range is tight and the norming point is not in the middle of the range as suggested by critical mass theory (Kanter, 1977).

5.7 Practical Implications

For policymakers, the results indicate that setting a minimum requirement, or making it more desirable, for diversity in boards is not enough to drive quality of reporting on sustainability; it seems that regulation of disclosure like the NFRD and CSRD – is doing much of the job within this sample. The findings indicate that corporate boards and executives are more likely to achieve positive reporting outcomes if they incorporate dedicated sustainability governance bodies (e.g., sustainability committees) and/or accountability for reporting (e.g., sustainability officer) into their reporting frameworks, or if they have reporting expertise on their boards (e.g., a sustainability specialist). The findings raise an investor and analyst alert that an investor cannot rely on board gender composition to signal report quality or sustainability reporting; instead, it is more likely to be a reliable indicator to assess report substance such as when third-party assurance is present and adverse impacts are disclosed.

5.8 Unexpected Findings

The last two results are interesting as exceptions to the theoretical account developed above. First, the important negative lagged BGD coefficient (Model 3) is counterintuitive in light of the general theoretical expectation of a positive, board-driven effect of BGD on reporting quality. This result is interpreted as inconclusive because of the low number of observations in this model and the very high level of multicollinearity among the governance variables and is thus highlighted as a priority for further replication in a larger sample before it may be considered theoretically important.

Second, Adverse Impact Disclosure (D6) was the least developed SRQ dimension along the observation period even as most other dimensions developed significantly. This persistent gap

that while disclosure breadth and framework adoption have improved, the willingness to report unfavorably has not suggests a selective disclosure that is not explained by the governance factors or the regulatory-timing argument put forward in this chapter, as the regulatory pressure appears to have had a positive effect on the willingness of firms to disclose broadly, but not unfavorably. This is observed here, and in subsequent chapters in this book (Chapter 6) as a pattern that is worthy of future research.

In addition, the low and unstable BGD coefficients that have been reported throughout this chapter should be interpreted with the critical mass theory caveat that BGD clusters very tightly around the threshold suggested by the theory, making it difficult to determine if the effect is really absent or if it is not being detected. The present study does not clarify this ambiguity and is intended to be continued as a limitation and a direction for future study in Chapter 6.

5.9 Chapter Summary

The findings in Chapter 4 have been read as most consistent with the legitimacy and institutional-theory explanation, namely that sustainability reporting quality increased significantly for the sample as a result of a shared regulatory pressure (coercive isomorphism); similarly, the change in board gender diversity over the same period did not show a strong, independent link to reporting quality once these common time trends are properly accounted for. Critical mass theory adds an important qualification: the linear specification of the model used in this study fits a small range of governance measures, which may lead to a misdetection of a threshold effect if one exists. In addition, the chapter points to methodological, theoretical, and practical implications of these findings and has highlighted two results that warrant careful evaluation and future research: (1) the negative lagged BGD coefficient and (2) the continued lack of adverse impact disclosure. These threads are brought together at the end of the study in Chapter 6 which leads to overall conclusions, contributions, limitations, and recommendations for future research.

Chapter 6: Conclusion

6.1 Research Summary

The aim of this study was to analyze the relationship between board gender diversity (BGD) and sustainability reporting quality (SRQ) for Swedish large cap industrial companies in the period 2016-2023, and to answer three research questions: 1) Is there a positive relationship between BGD and SRQ? 2) Do the dimensions of SRQ correlate with BGD? 3) How has this relationship changed as the sustainability disclosure regulation has tightened? The analysis employed a longitudinal document-based approach for the 15 firms, which resulted in 120 firm-year observations from annual and sustainability reports, from non-financial statements and from governance disclosures.

BGD was operationalized through multiple governance measures, including the proportion of female directors, board size, the Blau Diversity Index, and lagged BGD, consistent with recommendations in the governance-diversity literature to avoid relying on a single measure of board composition (Behlau, Bäumer, & Brem, 2024). SRQ was assessed across six dimensions framework adoption, target specificity, disclosure scope, materiality assessment, third-party assurance, and adverse-impact disclosure each scored from 0 to 3 using a structured content-analysis protocol, with inter-coder reliability formally assessed following established guidance for qualitative content analysis in business research (Coleman, Ragan, & Dari, 2024; Lim, 2024). The empirical analysis was carried out in four steps: descriptive statistics, correlation analysis, cross-firm and within-firm longitudinal comparison, and fixed-effects panel regression with firm and year controls. This design incorporates a structured content analysis and panel-based statistical evidence and was built throughout to maintain clear distinction between association and causation.

6.2 Key Findings

6.2.1 RQ1: Is board gender diversity positively associated with sustainability reporting quality?

The findings are not robust and independent and do not support the theory of a positive link between board gender diversity and how well the sustainability reports are produced. There was a weak, slightly negative bivariate correlation between the proportion of female directors and SRQ. The association is positive and statistically significant when only firm fixed effects are included in the regression, but when adding year fixed effects the association becomes

statistically insignificant, suggesting that the association may be substantial due to a shared time trend (most plausibly the stricter EU sustainability disclosure requirements over the period of observation) and not due to a firm-specific effect of board composition This pattern echoes a broader body of governance research in which apparent diversity effects on disclosure or performance outcomes weaken or disappear once appropriate controls are introduced (Sila, Gonzalez, & Hagendorff, 2016; Walls, Berrone, & Phan, 2012).

A second model, with the inclusion of lagged BGD, yielded a statistically significant negative value for BGD, which contradicts the theory and is viewed with skepticism because of the small sample of firms in this specification and the high degree of multicollinearity among the governance variables included.

Thus, the answer to RQ1 is properly guarded. There is no evidence that an increase in the number of female directors is associated with higher quality sustainability reporting, when shared regulatory pressure is controlled for. At the same time, the results cannot prove the complete absence of a relationship: as critical mass theory suggests (Kanter, 1977), the sample's board gender diversity clusters narrowly around the threshold at which minority board representation is theorized to become influential, and a linear model applied to this narrow range may simply be poorly positioned to detect a genuine non-linear effect, even if one exists in the underlying population of firms.

6.2.2 RQ2: Which dimensions of sustainability reporting quality vary most with board gender composition?

Consistent with prior evidence that sustainability reporting quality is a multidimensional rather than a unitary construct (Hahn & Kühnen, 2013; La Torre, Sabelfeld, Blomkvist, & Dumay, 2023), the six SRQ dimensions in this study showed clear differences in their development. The scope of disclosure and framework adoption showed the highest level of development and the most moderate level of development respectively during the observation period, with least development for adverse-impact disclosure, and the most variability across firm-years. In the dimension level, with firm effects and year effects added to the model, none of the individual dimensions were significantly associated with BGD and the direction of the coefficient was not the same across dimensions. This is not what a naive pattern would look like, where BGD coefficients may be positive and non-significant at the individual level, but rather the pattern obtained at the dimension level is a clear confirmation of the pattern obtained at the composite

level under RQ1: no strong and solid BGD effect is detectable after controlling for the shared time trends at the individual level.

6.2.3 RQ3: How does the relationship evolve between 2016 and 2023?

The longitudinal analysis reveal a significant decoupling of board gender diversity and sustainability reporting quality. SRQ also rose significantly and continuously over the years, more than tripling from a mean of 5.27 in 2016 to 17.54 in 2023, and the proportion of female directors increased by a mere 0.8% from 27.1% to 27.9% over the same period. The two dimensions that experienced the greatest improvement over time were materiality assessment and third-party assurance, which complemented the release of the EU Non-Financial Reporting Directive (NFRD) in 2017 for the first time in Sweden and the preparatory stage of the Corporate Sustainability Reporting Directive (CSRD) from 2021. This timing is consistent with a substantial body of evidence that mandatory and anticipated sustainability disclosure regulation is a primary driver of reporting improvement (Christensen, Hail, & Leuz, 2021; Pantazi, 2024).

The same trend is observed in the within-firm analysis. The proportion of female board members showed a small, almost negligible, change over time for almost all of the sampled firms, and for the sample as a whole, the SRQ increased over time. Although there was a clear case of a BGD change (from 27% to 36% females) and a corresponding increase in SRQ score among Atlas Copco, this case is not enough to draw broad conclusions about the correlation between BGD change and SRQ change.

6.3 Contributions of the Study

6.3.1 Methodological Contributions

This study makes a methodological contribution by implementing a systematical dimension-based content analysis along with a longitudinal firm-panel analysis. Rather than treating sustainability reporting as a single, broad measure of disclosure, as much of the existing literature does, this study distinguishes six dimensions covering framework adoption, target specificity, disclosure scope, materiality assessment, third-party assurance, and adverse-impact disclosure, following calls in the literature for more granular, multidimensional measurement of reporting quality (Hahn & Kühnen, 2013). The use of independent coders and formally assessed inter-coder reliability, consistent with established guidance for content-analysis research (Coleman, Ragan, & Dari, 2024; Lim, 2024), strengthens the comparability of SRQ

scores across firms and years. A second methodological implication of the empirical analysis is suggested by the fact that this study shows that associations between BGD–SRQ variables estimated without proper account for shared time trends may overestimate the extent to which board composition appears to matter. In this study, whether a model specification included year fixed effects or not, a positive BGD coefficient was statistically significant, but the effect disappeared when the model was more fully specified by controlling for the same set of regulatory pressures facing all firms in a given year. This is consistent with methodological concerns raised elsewhere about the complexity of isolating governance effects from concurrent regulatory change (Christensen, Hail, & Leuz, 2021), and suggests that future panel-based BGD–disclosure research spanning periods of regulatory transition should treat time-trend confounding as a central methodological concern rather than a secondary robustness check.

6.3.2 Empirical Contributions

Empirically, this study demonstrates that board gender diversity is not reliably associated with sustainability reporting quality once shared time trends are properly controlled for, extending a body of meta-analytic and firm-level evidence in which board composition effects on disclosure and CSR-related outcomes are often small, inconsistent, or contingent on model specification (Velte, 2022; Sila, Gonzalez, & Hagendorff, 2016).

The observation is reinforced by the panel structure of the analysis, which involves a 8-year time period instead of comparing a single cross section of firms. The research also shows empirical evidence of a differential rate of development across the dimensions of SRQ: firms are much further developed in terms of the scope of issues being disclosed and the frameworks they are using than they are in independent assurance and adverse impacts disclosure. This uneven development is itself an empirical contribution, since studies relying on a single aggregate disclosure score would not be able to observe it (La Torre, Sabelfeld, Blomkvist, & Dumay, 2023).

6.3.3 Contextual Contributions

This study provides context-specific evidence from Swedish large-cap industrial companies, where the level of board gender diversity is relatively high and stable, compared to many other international companies. This gives an in-depth description of the evolution of governance composition and sustainability reporting, along with any disconnections, in a specific

institutional and regulatory setting over a given time horizon. The finding that reporting maturity increased substantially while board gender composition remained largely stable is itself a contextual contribution, since it demonstrates that in a market where board gender norms already produce comparatively high baseline diversity, further variation in BGD may carry limited marginal explanatory power for reporting outcomes (Schoonjans, 2024).

6.3.4 Academic Contributions

In the academic sense, this study proves that the link between BGD and SRQ is not limited to a positive or negative characterization. Board gender composition is one facet of a firm's governance arrangements, but reporting outcomes also depend on organizational reporting structures, sustainability-specific capabilities, and external regulatory and normative demands (Lewellyn & Muller-Kahle, 2024; Recendes, Chandler, Huang, & Hill, 2024).

The study's theoretical framework, drawing on legitimacy theory, institutional theory, and critical mass theory, further supports this more contingent view. Legitimacy theory and institutional theory, in particular, offer the interpretation best supported by the data: firms appear to have converged on improved reporting practices in response to shared regulatory and normative pressure, consistent with coercive isomorphism (DiMaggio & Powell, 1983), rather than as a function of internal governance change. This is also consistent with the distinction in the legitimacy literature between symbolic and substantive disclosure practices (Bota-Avram, 2025a): the finding that adverse-impact disclosure remains the weakest SRQ dimension even as other dimensions matured suggests that firms have been more willing to expand the appearance of comprehensive reporting than to substantively disclose unfavorable outcomes. Critical mass theory (Kanter, 1977) contributes a distinct academic caution rather than a competing explanation, namely that a linear empirical specification, applied to a sample in which BGD varies within a narrow range close to the theory's proposed threshold, may be unable to detect a genuine non-linear relationship even where one exists. Finally, the study underscores the need to consider the disclosure of SRQ as being measured across different dimensions (disclosure breadth, target specificity, materiality, assurance, and adverse-impact reporting) rather than summarizing disclosure into a single measure.

The results also feed into a larger debate on the relationship between corporate governance reforms and the United Nations Sustainable Development Goals (SDGs), and to SDG 5 (Gender Equality), which explicitly calls for women's full and effective participation and equal opportunities for leadership in public and private decision-making at all levels. Gender

diversity on the board is often promoted in policies and in corporate communication as a way of achieving the desired outcomes for SDG 5 through sustainability. Reporting quality is only partly corroborated by the results of this study: That BGD seem to be linked to sustainability reporting is mainly explained by the time effects of regulations rather than a stable, direct effect at the firm level. This pattern is consistent with concerns raised in the greenwashing literature that corporate diversity and sustainability commitments can function primarily as symbolic compliance signalling alignment with external expectations and audit-style scrutiny rather than as substantive drivers of organizational change (Lyon & Maxwell, 2011; Marquis, Toffel, & Zhou, 2016; Testa, Bosman, & Iraldo, 2025). This should not be interpreted as evidence that the sampled firms are actively greenwashing their boards; the design of this study does not allow for the observation of firms' intent and therefore interpretive conclusions should not be drawn about this characteristic. This should be read as a sign that, at least in this sample and time, greater board gender diversity has not led to a measurable improvement in the depth or candour of sustainability reporting; the latter is the economically more significant measure that is often assumed as the outcome of SDG 5-related governance reforms.

6.4 Practical Implications

The results indicate that solely appointing more women to the board's director roles is not enough to achieve sustainable improvement in sustainability reporting practices. Proper governance of the reporting process itself, together with relevant sustainability-reporting experience and expertise on the board, appears more directly connected to reporting quality than board gender composition on its own (Lewellyn & Muller-Kahle, 2024). For corporate governance arrangements more broadly, it is important to look beyond diversity metrics and to consider the knowledge, experience, and functional capabilities represented on the board, consistent with a resource-oriented view of what boards contribute to organizational outcomes (Recendes, Chandler, Huang, & Hill, 2024).

The findings suggest a focus for sustainability reporting teams on the quality and authenticity of the reporting process rather than just on the extent of reporting. This will involve establishing clear and measurable objectives, incorporating a strong materiality assessment framework and seeking third-party assurance, as well as upgrading disclosures on adverse impacts specifically, where there are some weaker elements within the sample but overall this was the area where there was most improvement. The findings indicate that board gender diversity should not serve

as a proxy for sustainability reporting quality and that investors and analysts should consider using other metrics instead. Rather, investors should examine in detail whether the sustainability practices being reported are measurable, independently verified, and whether they are delivering a balanced approach to both the good and the bad outcomes of sustainability.

For policymakers, the findings support continued emphasis on strengthening disclosure regulation itself, including specific attention to assurance requirements and adverse-impact disclosure, since these appear to be the areas where voluntary improvement has lagged furthest behind other aspects of reporting, even as overall reporting quality has advanced markedly under regulatory pressure such as the NFRD and the CSRD (Pantazi, 2024). More broadly, the findings caution against treating board gender diversity targets as a sufficient proxy for progress on SDG 5 (Gender Equality) at the corporate level: if board diversity quotas or targets are pursued as a compliance exercise disconnected from the substance of sustainability reporting and organizational practice, they risk functioning as a symbolic signal rather than a driver of the economically meaningful change that SDG-aligned governance reform is intended to produce (Lyon & Maxwell, 2011; Marquis, Toffel, & Zhou, 2016).

6.5 Limitations

The present study has some limitations that need to be considered when analyzing the results. First, the sample includes only 15 firms, resulting in just 120 firm-year observations, but even with the absence of financial and governance data (between 62 and 71 observations across between 9 and 10 firms), the regression models have relatively low statistical power, and the estimated coefficients are relatively imprecise, especially for the lagged-BGD specification.

Second, the study focuses on a specific group of companies: large-cap industrial companies in Sweden, and any conclusions about their findings should not be extended to other contexts, such as smaller companies, service-based companies, or companies in markets with significantly different regulatory structures or levels of gender diversity in the boardroom. Third, elements of judgement remain inherent to content analysis. Although inter-coder reliability testing was conducted and a coding protocol was pre-established, achieving consistently high agreement ($\kappa = 0.70–0.85$ across dimensions), some categories of disclosure, adverse-impact disclosure in particular, are inherently more difficult to code consistently than others (Coleman, Ragan, & Dari, 2024). The reliance on publicly disclosed information is itself a further constraint: the study measures what firms choose to disclose rather than what occurs internally, and cannot observe governance processes or board deliberations directly.

Fourth, the eight-year annual panel prevents the study from examining longer-term links, and an annual observation cycle might not be able to fully capture the lag between a change in the composition of the board and a change in the quality of reporting. The three governance variables employed (percentage of female directors, Blau Diversity Index and lagged BGD) are also extremely strongly correlated ($r = 0.95–0.99$), constraining the possibility of dissociating them empirically and possibly explaining the volatility of the lagged-BGD coefficient found under RQ1.

Fifth, the linear specification that has been used in all the regression analyses assumes that the marginal effect of BGD is constant over the entire range of BGD.As discussed in Chapter 5, critical mass theory (Kanter, 1977) predicts that minority board representation may influence organizational outcomes non-linearly, with effects becoming detectable only beyond a critical threshold commonly proposed at approximately 30 percent representation. The linear specification employed here is rather narrow in its cluster around BGD, between 22 and 36 percent, and this may well not be well adapted to uncovering any true non-linear relation in the population of firms, regardless of the actual presence of such a relation in the population.

Lastly, as in all observational panel designs, this study cannot claim a causal link between board gender diversity and sustainability reporting quality, and some of the patterns in the relationships examined may be due to variables unobserved in the model, such as board members' sustainability-specific expertise, the fact that the firms had a dedicated sustainability committee, or the firms' sustainability-related strategic priorities.

6.6 Recommendations for Future Research

A larger, cross-country comparison could be used to expand this study and determine if the pattern found in this study a significant increase in reporting quality but little change in the baseline level of board diversity is unique to the Swedish institutional setting or if it is the norm in markets across Europe with varying institutional structures and levels of board diversity.

Further research should also include non-gender-related governance factors including directors' sustainability-related expertise, the existence of a sustainability committee with a specific mandate, board independence, director tenure, international experience and background diversity of directors. This would allow more direct empirical tests of board-level mechanisms such as resource dependence and upper echelons effects, which lie outside the theoretical scope

of the present study, than gender representation alone can provide (Recendes, Chandler, Huang, & Hill, 2024; Lewellyn & Muller-Kahle, 2024).

From methodological perspective, future research should test non-linear or threshold-based specifications of the BGD–SRQ relationship, for example, spline functions or categorical threshold indicators around the critical mass level proposed by Kanter (1977) ideally applied to a sample with wider variation in board gender composition than was available in this study, in order to determine whether the near-null linear result reported here reflects a genuine absence of effect or a specification unable to detect a threshold-based one. The use of machine-assisted or computational content analysis represents a further promising direction, potentially allowing much larger collections of sustainability reports to be analyzed while retaining the same six-dimension classification scheme used in this study, consistent with recent methodological advances in automated content-analysis research (Lim, 2024; Testa, Bosman, & Iraldo, 2025).

Given that this study's observation period concludes as the CSRD moves from a preparatory to a mandatory phase, a valuable extension would be to examine sustainability reporting quality specifically through the period of CSRD implementation, to assess directly whether intensified regulatory requirements produce further, measurable improvement in reporting quality, including in the weaker dimensions identified here, such as adverse-impact disclosure and third-party assurance (Christensen, Hail, & Leuz, 2021; Pantazi, 2024).

Finally, future research could expand the analyses to other industries; use a broader array of diversity measures; employ larger European panels; and may wish to delve deeper into board decision making processes and discussions by conducting interviews with directors instead of relying on public disclosures, which would enable the study to capture the internal governance process and deliberation that is not evident in the public disclosures.

6.7 Concluding Remarks

The aim of this study was to investigate whether board gender diversity is correlated with the quality of sustainability reporting in Swedish large-cap industrial companies from 2016 to 2023 and the direction of the relationship. Once these shared time trends in regulation are controlled for, there is no strong statistically significant relationship between the evidence. However, there is also clear evidence that the quality of sustainability reporting has improved significantly during the observation period, but gender diversity of the boards has not.

The study nevertheless has value in showing that sustainability reporting practice is multidimensional and not well developed the extent of disclosure and use of sustainability reporting frameworks are relatively well developed practices among the companies studied, whereas independent assurance, especially disclosure on adverse impact, are still relatively immature practices. It also demonstrates that substantial improvement in reporting quality can occur without corresponding change in board gender composition, a pattern most consistent with legitimacy theory and institutional theory specifically, firms converging on improved practice in response to shared regulatory pressure such as the NFRD and CSRD, rather than as a function of internal governance change (DiMaggio & Powell, 1983; Bota-Avram, 2025a).

Overall, the results of this study indicate that the issue of board gender diversity must be interpreted in the context of a firm's overall governance and reporting practices and is not sufficient to explain the quality of sustainability reporting. In the face of this wider policy discourse that correlates board gender diversity to SDG 5 and to the wider corporate sustainability performance, these conclusions are a word of caution: if there is no real and clear connection to the reporting substance, board diversity targets can be pursued, but they may not provide the economic or organizational change that is intended. The study contributes to a dimension-based, systematic measurement of reporting quality, to a 8-year longitudinal panel data, and to a comparison of firm-level reporting quality in a specific and well-defined institutional context. Given the small sample size, the fact that all of the governance measures are highly correlated with one another, and the non-linear nature of the relationship between board gender diversity and sustainability reporting quality as suggested by critical mass theory, any conclusions drawn from its results should not be interpreted as causal, and the actual relationship between board gender diversity and sustainability reporting quality is truly context-dependent and an open question for future research.

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